CATEGORIES OF OPTIMIZATION REDUCTIONS

A. Sepúlveda-Jiménez[1,2]

[1]National University
[2]QDR Labs

January 10, 2026

ABSTRACT

We develop a categorical semantics for "reduce–solve–recover" pipelines in optimization. Objects are optimization problems presented as costed sets, while morphisms are certified (exact) reductions equipped with solution recovery. Several standard sets associated to a problem—candidate points, feasible points, and global minimizers—assemble into functors, and basic relations between them become natural transformations. Objective-value guarantees can be expressed as (op)lax naturality conditions in a poset-enriched codomain. We then generalize deterministic maps to nondeterministic or randomized maps by working in a Kleisli category of a monad, so that many constructions become statements about natural transformations "in the Kleisli world" [45, 70]. We then extend the framework in two practically important directions: (i) approximation, where reductions are allowed multiplicative and additive slack; and (ii) resources, where morphisms carry runtime/memory costs tracked by monoidal/enriched structure. W also generalize this categorification of optimization reductions to stochastic fields. We integrate these optimization reduction categories with game-theoretic categorifications. Finally, we present preliminary approaches to some open problems left.

Contents

1 Introduction **4**

2 Preliminaries **5**

2.1 Extended-real objectives 5

2.2 Categories, functors, and natural transformations 5

3 A category of exact optimization reductions **5**

3.1 Morphisms as certified reductions with recovery 5

3.2 Category structure 5

3.3 Basic consequences 6

4 Functorial constructions from $\mathbf{OptRed}_{\mathrm{ex}}$ **6**

4.1 Candidate, feasible, and solution functors 6

4.2 Contravariant variants via decoding 6

5 Natural transformations in this scheme **6**

5.1 Canonical inclusions 6

5.2 Contravariant inclusions on $\mathbf{OptRed}_{\mathrm{ex}}^{op}$ . . . 7
5.3 Natural transformations as representation-invariant post-processing . . . 7
5.4 Reductions as natural transformations via Yoneda . . . 7

6 Costs as oplax natural transformations in a poset-enriched setting 8
6.1 The 2-category $\mathbf{Pos}$ and discrete embedding . . . 8
6.2 An oplax naturality statement . . . 8

7 Kleisli generalization: randomness and nondeterminism 8
7.1 Monads and Kleisli categories . . . 8
7.2 Cost algebras and Kleisli lifting . . . 8
7.3 A category of T-effectful optimization reductions . . . 9
7.4 Kleisli-valued functors and natural transformations . . . 9
7.5 Naturality in the Kleisli world . . . 9
7.6 Costs as oplax naturality via Kleisli lifting . . . 10

8 Approximate reductions 10
8.1 Affine slack parameters as a monoid . . . 10
8.2 Deterministic approximate reductions . . . 10
8.3 Approximation in the Kleisli world . . . 11

9 Resources and enrichment 11
9.1 A resource monoidal preorder . . . 12
9.2 Resource-annotated reductions . . . 12
9.3 Enrichment viewpoint: minimal resource as a Lawvere metric . . . 12
9.4 Internalizing resources via a writer monad and transformer . . . 13

10 A unified graded/effectful structure 13
10.1 The product grade monoid $\mathbf{Aff}_{\geq 0} \times W$ . . . 13
10.2 Externally graded effectful reductions . . . 13
10.3 Internalizing the grade via a writer transformer . . . 14

11 Application instantiations 14
11.1 Convex optimization as a structured subcategory . . . 15
11.2 Nonconvex optimization and relax–round pipelines . . . 15
11.3 Linear programming and the simplex family . . . 15
11.4 Stochastic optimization and stochastic programming . . . 15
11.5 Heuristic and evolutionary optimization . . . 16

12 Optimization transformations for ensemble and averaging methods in machine learning 16
12.1 Supervised learning objects and ensemble decision spaces . . . 16
12.2 Bagging, random forests, and bootstrap aggregation as Kleisli morphisms . . . 17

12.3 Boosting as optimization over convex hulls and stagewise composition . . . 17
12.4 Deep ensembles, checkpoint averaging, and weight-space averaging . . . 17

13 Categorifying game theory and integrating optimization reductions 18
13.1 Normal-form games in cost form . . . 18
13.2 A Nash-gap functional and reduction to single-objective optimization . . . 18
13.3 A category of game reductions compatible with $\mathbf{OptRed}^{\mathrm{gr}}_{T,\alpha}$ . . . 19
13.4 Special classes: potential games, zero-sum games, and convex (Rosen) games . . . 19
13.5 Randomization: mixed strategies, correlated equilibrium, and stochastic games . . . 20
13.6 Compositional game theory and open games . . . 20

14 Random fields and general stochastic fields 21
14.1 Random fields as random variables in configuration spaces . . . 21
14.2 Structured subcategories: Gaussian fields and Markov random fields . . . 21
14.3 Field-valued optimization problems and risk aggregation . . . 22
14.4 Discretization, coarsening, and multilevel reductions . . . 22
14.5 Stochastic field dynamics and inference kernels . . . 22

15 Higher-categorical extensions: from $\mathbf{OptRed}^{\mathrm{gr}}_{T,\alpha}$ to n-categories 22
15.1 A minimal locally posetal 2-category of reductions . . . 23
15.2 Proof-relevant 2-cells and bicategorical semantics . . . 23
15.3 From bicategories to n-categories . . . 24
15.4 Higher natural transformations . . . 24
15.5 Higher Kleisli objects and graded effects . . . 24

16 Infinity-categorical extensions 25
16.1 Relative categories of reductions and simplicial localization . . . 25
16.2 Functorial constructions and naturality in ∞-categories . . . 25
16.3 Effect monads, Kleisli ∞-categories, and graded effects . . . 26
16.4 Remark: beyond $(\infty, 1)$ . . . 26

17 (∞, n)-categorical extensions and canonical weak equivalences 26
17.1 From relative n-categories of reductions to an (∞, n)-category . . . 26
17.2 Canonical choices of W in three domains . . . 27
17.2.1 (i) Randomness: $\mathcal{D}$ and kernel–coupling equivalences . . . 27
17.2.2 (ii) Nondeterminism: $\mathcal{P}$ and relational isomorphisms . . . 28
17.2.3 (iii) Syntactic algorithms: certified compiler transformations . . . 28

18 Conclusion and outlook 28

19 Preliminary Approaches to Open Problems 29
19.1 Non-invertible Higher Structure . . . 29

19.2 Duality and Relaxation Theory 29
19.3 Mechanization and Verification 29
19.4 Applications in Stochastic Fields 30

1 Introduction

Optimization problems rarely appear in isolation. In practice one typically builds a *pipeline—reduce/preprocess* $\to$ *solve* $\to$ *recover/postprocess*—that transforms an input instance into a form amenable to a chosen solver, then translates the solver output back into a solution for the original problem. This is pervasive across operations research and convex optimization [64, 9], as well as theoretical computer science where reductions formalize relations among problems and complexity classes [22, 57]. The same pattern appears in modern optimization toolchains: presolve and scaling in LP/MIP, relaxations and rounding in approximation algorithms, and surrogate objectives and stochastic estimators in machine learning.

Category theory is a natural fit for reasoning about such pipelines, because it is designed to make *compositional* structure explicit: objects, morphisms, and the laws governing their composition [44, 4]. The guiding idea of this paper is to treat optimization problems as objects and certified "reduce–solve–recover" procedures as morphisms, so that chaining procedures is just categorical composition. Once one has a category of problems and transformations, a large fraction of the structural statements one informally uses in optimization ("feasibility is preserved by preprocessing", "solutions map back along recovery maps", "bounds compose multiplicatively/additively", etc.) become functorial or naturality statements.

From deterministic reductions to effectful pipelines. A clean deterministic starting point is to encode constraints via *extended-real* objectives: a problem is a set X of decisions together with a cost $c : X \to \overline{\mathbb{R}}$, with infeasible points assigned $+\infty$ [64]. A certified reduction $P \to Q$ then consists of an encoding map and a decoding map whose costs are monotone in the appropriate directions. However, many real algorithms are not deterministic total functions: they are nondeterministic (branch-and-bound search, local improvement), randomized (stochastic gradient methods), or partial (timeouts, numerical failure). To capture this *uniformly*, we replace $\mathbf{Set}$-valued semantics by the *Kleisli category* of a monad T [45, 70]. Nondeterminism can be modeled by the powerset monad, while randomness can be modeled by probability distribution monads such as Giry's [24, 30]. Many of the deterministic constructions then reappear as *Kleisli-natural* transformations.

Approximation and resources as first-class structure. Exact reductions are too strict for approximation algorithms and heuristics. We therefore weaken correctness to allow multiplicative and additive slack (e.g. $c_P(\mathrm{dec}(y)) \leq \alpha\, c_Q(y) + \beta$), recovering the familiar systematic composition law: multiplicative factors multiply while additive terms accumulate (with upstream scaling). Separately, pipelines must be assessed not only by solution quality but also by resource usage (runtime, memory). We incorporate resources by enriching morphisms with a monoidal "cost" and using monoidal/enriched structure to propagate resource bounds through composition [33, 37].

A unified graded/effectful view. To track *effects* (randomness/nondeterminism), *approximation* loss, and *resources* in one compositional semantics, we introduce a single grade monoid $G = \mathbf{Aff}_{\geq 0} \times W$ whose composition law is the simultaneous bookkeeping rule for quality and resources. This can be treated externally (as morphisms annotated by grades) or internalized by a writer-style transformer over the effect monad, in the spirit of graded and parameterized effects [2, 32, 55].

Why higher categories. Once algorithms are morphisms, it is natural to ask what should count as "the same" algorithm. In practice we routinely identify pipelines up to semantics-preserving rewrites (algebraic simplifications, compiler optimizations, alternative factorizations, coupling arguments, . . .). These are not mere equalities of functions: they have *structure* and *coherence*. Higher category theory is the natural language for this: 2-morphisms witness transformations between morphisms [6], n-categories organize higher coherences [38], and ∞-categorical localization turns a category with weak equivalences into a homotopy-coherent world of mapping spaces [14, 59, 42]. In particular, an (∞, n)-framework allows one to treat invertible higher coherences above level n while keeping non-invertible refinement or optimization witnesses at lower dimensions [5, 60].

Canonical weak equivalences in three domains. To make the higher-categorical constructions concrete, we single out canonical choices of weak equivalences W in three settings that arise in optimization pipelines: (i) for probabilistic effects ($T = \mathcal{D}$), we use equivalences generated by *kernel couplings* and measurable isomorphisms, as in

probabilistic coupling theory [41, 21]; (ii) for nondeterminism ($T = \mathcal{P}$), we use *relational isomorphisms* in the sense of allegories/relations [19]; and (iii) for syntactic algorithm languages, we take W to be *certified compiler transformations* (semantics-preserving rewrites equipped with proof objects), following the verified-compilation ethos [39]. The resulting (∞, n)-mapping objects can be read as "spaces" of pipelines together with all higher rewrite coherences.

Roadmap. Sections 1–5 develop the deterministic base category of exact reductions and its functorial/natural structure. Sections 6–9 extend the setting to Kleisli effects, approximation slack, resources, and the unified graded/effectful framework. Sections 10–12 then lift the semantics to higher categorical levels, culminating in an (∞, n)-localization equipped with canonical weak equivalences for randomness, nondeterminism, and certified syntactic transformations.

2 Preliminaries

2.1 Extended-real objectives

Let $\overline{\mathbb{R}} := \mathbb{R} \cup \{+\infty\}$ with the usual total order extended by $x < \infty$ for all $x \in \mathbb{R}$. We encode hard constraints by assigning cost $+\infty$ to infeasible points [64, 9].

Definition 2.1 (Optimization problem as a costed set)**.** An *(minimization) optimization problem* is a pair

$$P = (X, c_P),$$

where X is a set (the decision set) and $c_P : X \to \overline{\mathbb{R}}$ is an extended-real objective [64, 9]. Define

$$\mathsf{Feas}(P) := \{x \in X : \ c_P(x) < \infty\}, \qquad \mathrm{val}(P) := \inf_{x \in X} c_P(x),$$

and the (possibly empty) set of global minimizers

$$\mathsf{Sol}(P) := \arg\min_{x \in X} c_P(x) = \{x \in X : \ c_P(x) = \mathrm{val}(P)\}.$$

2.2 Categories, functors, and natural transformations

We assume standard category theory [44, 4]. We write $\mathbf{Set}$ for sets and functions, and $\mathbf{Pos}$ for the (2-)category of posets and monotone maps, regarded as a locally posetal 2-category where 2-cells are pointwise inequalities. Natural transformations are as usual for $\mathbf{Set}$-valued functors; for $\mathbf{Pos}$-valued functors we will also use (op)lax natural transformations.

3 A category of exact optimization reductions

3.1 Morphisms as certified reductions with recovery

Intuitively, a reduction $a : P \to Q$ is an algorithm that transforms instances of P into instances of Q and can recover (optimal) solutions for P from solutions for Q; compare standard reduction paradigms in complexity theory [22, 57].

Definition 3.1 (Exact reduction morphism)**.** Let $P = (X, c_P)$ and $Q = (Y, c_Q)$. An *exact reduction morphism* $a : P \to Q$ is a pair of functions

$$a = (\mathrm{enc}_a : X \to Y, \ \mathrm{dec}_a : Y \to X)$$

such that for all $x \in X$ and $y \in Y$,

$$c_Q(\mathrm{enc}_a(x)) \le c_P(x), \tag{1}$$
$$c_P(\mathrm{dec}_a(y)) \le c_Q(y). \tag{2}$$

3.2 Category structure

Proposition 3.2 (Category structure)**.** *Exact reductions form a category* $\mathbf{OptRed}_{\mathrm{ex}}$ *with:*

- *objects: optimization problems* (X, c)*,*
- *morphisms: exact reductions* $a = (\mathrm{enc}_a, \mathrm{dec}_a)$*,*
- *identity on* $P = (X, c_P)$*:* $\mathrm{id}_P = (\mathrm{id}_X, \mathrm{id}_X)$*,*

- *composition: for $a : P \to Q$ and $b : Q \to R$,*

$$(b \circ a) := (\mathrm{enc}_b \circ \mathrm{enc}_a,\ \mathrm{dec}_a \circ \mathrm{dec}_b).$$

Proof. Identity laws are immediate. For composition, the encoding inequality follows from

$$c_R(\mathrm{enc}_b(\mathrm{enc}_a(x))) \le c_Q(\mathrm{enc}_a(x)) \le c_P(x),$$

and the decoding inequality follows from

$$c_P(\mathrm{dec}_a(\mathrm{dec}_b(z))) \le c_Q(\mathrm{dec}_b(z)) \le c_R(z).$$

Associativity holds because it holds for function composition. □

3.3 Basic consequences

Lemma 3.3 (Feasibility and optimal values)**.** *If $a : P \to Q$ is an exact reduction, then $\mathrm{enc}_a(\mathsf{Feas}(P)) \subseteq \mathsf{Feas}(Q)$ and $\mathrm{dec}_a(\mathsf{Feas}(Q)) \subseteq \mathsf{Feas}(P)$. Moreover $\mathrm{val}(P) = \mathrm{val}(Q)$.*

Proof. If $x \in \mathsf{Feas}(P)$ then $c_P(x) < \infty$, hence $c_Q(\mathrm{enc}_a(x)) \le c_P(x) < \infty$, so $\mathrm{enc}_a(x) \in \mathsf{Feas}(Q)$. Similarly, if $y \in \mathsf{Feas}(Q)$ then $c_P(\mathrm{dec}_a(y)) \le c_Q(y) < \infty$, so $\mathrm{dec}_a(y) \in \mathsf{Feas}(P)$.

Taking infima in (1) gives $\mathrm{val}(Q) \le \mathrm{val}(P)$, and taking infima in (2) gives $\mathrm{val}(P) \le \mathrm{val}(Q)$. □

Lemma 3.4 (Minimizers map back)**.** *If $a : P \to Q$ is an exact reduction and $y^\star \in \mathsf{Sol}(Q)$, then $\mathrm{dec}_a(y^\star) \in \mathsf{Sol}(P)$.*

Proof. By (2), $c_P(\mathrm{dec}_a(y^\star)) \le c_Q(y^\star) = \mathrm{val}(Q) = \mathrm{val}(P)$ (Lemma 3.3), so $\mathrm{dec}_a(y^\star)$ attains $\mathrm{val}(P)$. □

4 Functorial constructions from $\mathbf{OptRed}_{\mathrm{ex}}$

4.1 Candidate, feasible, and solution functors

Definition 4.1 (Candidate functor)**.** Define $\mathsf{Cand} : \mathbf{OptRed}_{\mathrm{ex}} \to \mathbf{Set}$ by $\mathsf{Cand}(P) = X$ for $P = (X, c_P)$ and $\mathsf{Cand}(a) = \mathrm{enc}_a$ for $a : P \to Q$.

Definition 4.2 (Feasible-set functor)**.** Define $\mathsf{Feas} : \mathbf{OptRed}_{\mathrm{ex}} \to \mathbf{Set}$ by $\mathsf{Feas}(P) = \{x \in X : c_P(x) < \infty\}$ and $\mathsf{Feas}(a) = \mathrm{enc}_a|_{\mathsf{Feas}(P)}$.

Definition 4.3 (Solution functor on $\mathbf{OptRed}_{\mathrm{ex}}^{op}$)**.** Define $\mathsf{Sol} : \mathbf{OptRed}_{\mathrm{ex}}^{op} \to \mathbf{Set}$ by $\mathsf{Sol}(P) = \arg\min c_P$ and for $a : P \to Q$,

$$\mathsf{Sol}(a^{op}) := \mathrm{dec}_a|_{\mathsf{Sol}(Q)} : \mathsf{Sol}(Q) \to \mathsf{Sol}(P).$$

This is well-defined by Lemma 3.4.

Proposition 4.4 (Functoriality)**.** Cand *and* Feas *are covariant functors* $\mathbf{OptRed}_{\mathrm{ex}} \to \mathbf{Set}$*, and* Sol *is a contravariant functor* $\mathbf{OptRed}_{\mathrm{ex}}^{op} \to \mathbf{Set}$.

Proof. For Cand and Feas, functoriality follows from $\mathrm{enc}_{b\circ a} = \mathrm{enc}_b \circ \mathrm{enc}_a$ and feasibility preservation (Lemma 3.3). For Sol, note that $\mathrm{dec}_{b\circ a} = \mathrm{dec}_a \circ \mathrm{dec}_b$ and restriction to $\mathsf{Sol}(R)$ yields $\mathsf{Sol}((b \circ a)^{op}) = \mathsf{Sol}(a^{op}) \circ \mathsf{Sol}(b^{op})$. □

4.2 Contravariant variants via decoding

Definition 4.5 (Decoding functors on $\mathbf{OptRed}_{\mathrm{ex}}^{op}$)**.** Define $\mathsf{Cand}^{op}, \mathsf{Feas}^{op} : \mathbf{OptRed}_{\mathrm{ex}}^{op} \to \mathbf{Set}$ by $\mathsf{Cand}^{op}(P) = X$ and $\mathsf{Cand}^{op}(a^{op}) = \mathrm{dec}_a$, and similarly $\mathsf{Feas}^{op}(a^{op}) = \mathrm{dec}_a|_{\mathsf{Feas}(Q)} : \mathsf{Feas}(Q) \to \mathsf{Feas}(P)$.

5 Natural transformations in this scheme

5.1 Canonical inclusions

Theorem 5.1 (Inclusions $\mathsf{Feas} \Rightarrow \mathsf{Cand}$ and $\mathsf{Sol} \Rightarrow \mathsf{Feas}$)**.** *There are natural transformations*

$$\iota : \mathsf{Feas} \Rightarrow \mathsf{Cand} \quad \textit{and} \quad \kappa : \mathsf{Sol} \Rightarrow \mathsf{Feas}^{op}$$

whose components are the set inclusions $\iota_P : \mathsf{Feas}(P) \hookrightarrow \mathsf{Cand}(P)$ and $\kappa_P : \mathsf{Sol}(P) \hookrightarrow \mathsf{Feas}(P)$.

Proof. For ι, naturality is the commutativity of

$$\begin{tikzcd} \mathsf{Feas}(P) \arrow[r, "\mathsf{Feas}(a)"] \arrow[d, hook, "\iota_P"'] & \mathsf{Feas}(Q) \arrow[d, hook, "\iota_Q"] \\ \mathsf{Cand}(P) \arrow[r, "\mathsf{Cand}(a)"'] & \mathsf{Cand}(Q) \end{tikzcd}$$

since both routes are enc_a with appropriate domains. For κ, if $a : P \to Q$, naturality on $a^{op} : Q \to P$ is the commutativity of

$$\begin{tikzcd} \mathsf{Sol}(Q) \arrow[r, "\mathsf{Sol}(a^{op})"] \arrow[d, hook, "\kappa_Q"'] & \mathsf{Sol}(P) \arrow[d, hook, "\kappa_P"] \\ \mathsf{Feas}(Q) \arrow[r, "\mathsf{Feas}^{op}(a^{op})"'] & \mathsf{Feas}(P) \end{tikzcd}$$

which holds because both routes are dec_a restricted, and $\mathrm{dec}_a(\mathsf{Sol}(Q)) \subseteq \mathsf{Sol}(P)$ (Lemma 3.4). □

5.2 Contravariant inclusions on $\mathbf{OptRed}_{\mathrm{ex}}^{op}$

Theorem 5.2 (Inclusions on $\mathbf{OptRed}_{\mathrm{ex}}^{op}$)**.** *There is a natural transformation $\iota^{op} : \mathsf{Feas}^{op} \Rightarrow \mathsf{Cand}^{op}$ given by inclusions $\mathsf{Feas}(P) \hookrightarrow X$ on each object P.*

Proof. The naturality square for $a^{op} : Q \to P$ commutes because both routes are the restriction of dec_a. □

5.3 Natural transformations as representation-invariant post-processing

A common optimization practice is to apply a *post-processing* map $h_P : X \to X$ that improves or simplifies a candidate solution without changing the underlying problem (e.g. removing redundant slack variables). In our setting, such a family is naturally a natural endotransformation of Cand if it is invariant under change of representation.

Definition 5.3 (Reduction-invariant post-processing)**.** A family $h = (h_P)_{P \in \mathrm{Ob}(\mathbf{OptRed}_{\mathrm{ex}})}$ with $h_P : \mathsf{Cand}(P) \to \mathsf{Cand}(P)$ is *reduction-invariant* if for every reduction $a : P \to Q$,

$$\mathrm{enc}_a \circ h_P = h_Q \circ \mathrm{enc}_a. \tag{3}$$

Proposition 5.4. *Reduction-invariant post-processing families are precisely natural transformations $h : \mathsf{Cand} \Rightarrow \mathsf{Cand}$.*

Proof. Naturality of h at $a : P \to Q$ is exactly (3). □

Example 5.5 (Slack-variable removal)**.** Let Q be an LP in standard form with slack variables, and P the corresponding inequality form. The map that projects $(x, s) \mapsto x$ is a post-processing step that is representation-invariant with respect to the standard-form reduction; hence it yields a natural transformation on the relevant subcategory of LP problems.

5.4 Reductions as natural transformations via Yoneda

Fix $P \in \mathbf{OptRed}_{\mathrm{ex}}$. The *representable* functor $\mathbf{OptRed}_{\mathrm{ex}}(-, P) : \mathbf{OptRed}_{\mathrm{ex}}^{op} \to \mathbf{Set}$ sends Q to the set of pipelines (reductions) from Q to P.

Definition 5.6 (Representable (pipeline) functor)**.** For $P \in \mathbf{OptRed}_{\mathrm{ex}}$, define $y(P) := \mathbf{OptRed}_{\mathrm{ex}}(-, P) : \mathbf{OptRed}_{\mathrm{ex}}^{op} \to \mathbf{Set}$.

Theorem 5.7 (Yoneda viewpoint)**.** *For any $P, Q \in \mathbf{OptRed}_{\mathrm{ex}}$, there is a natural bijection*

$$\mathrm{Nat}(y(P), y(Q)) \;\cong\; \mathbf{OptRed}_{\mathrm{ex}}(Q, P).$$

In particular, a reduction $r : Q \to P$ corresponds to precomposition by r, i.e. the natural transformation $\mathbf{OptRed}_{\mathrm{ex}}(-, Q) \Rightarrow \mathbf{OptRed}_{\mathrm{ex}}(-, P)$ whose component at R maps $f : R \to Q$ to $r \circ f : R \to P$.

Proof. This is the Yoneda lemma [44, 4]. □

6 Costs as oplax natural transformations in a poset-enriched setting

6.1 The 2-category $\mathbf{Pos}$ and discrete embedding

Recall that $\mathbf{Pos}$ is a locally posetal 2-category: for monotone maps $u, v : A \to B$ there is at most one 2-cell $u \Rightarrow v$, which exists exactly when $u(a) \leq v(a)$ for all $a \in A$ [33, 44].

6.2 An oplax naturality statement

Let $\overline{\mathbb{R}}$ be viewed as a poset and hence an object of $\mathbf{Pos}$.

Definition 6.1 (Poset-valued functor of cost functions)**.** Define $\mathcal{E} : \mathbf{Set}^{op} \to \mathbf{Pos}$ by

$$\mathcal{E}(X) := (\overline{\mathbb{R}})^X$$

with pointwise order, and for $f : X \to Y$ define $\mathcal{E}(f) : (\overline{\mathbb{R}})^Y \to (\overline{\mathbb{R}})^X$ by precomposition $\varphi \mapsto \varphi \circ f$.

Proposition 6.2 (Costs are oplax natural)**.** *Let $C := \mathcal{E} \circ \mathsf{Cand}^{op} : \mathbf{OptRed}_{\mathrm{ex}}^{op} \to \mathbf{Pos}$. Then the family of objective functions $(c_P)_{P \in \mathbf{OptRed}_{\mathrm{ex}}}$ is an* oplax natural transformation *from the constant functor $\Delta(\top)$ to C, where $\top$ is a singleton poset: for each $a : P \to Q$ there is a 2-cell (inequality)*

$$C(a^{op})(c_P) = c_P \circ \mathrm{dec}_a \ \leq \ c_Q.$$

Similarly, using Cand instead of Cand^{op} yields $c_Q \circ \mathrm{enc}_a \leq c_P$.

Proof. The inequalities are exactly (2) and (1) written as 2-cells in $\mathbf{Pos}$. Oplax naturality coherence is automatic in a locally posetal 2-category. □

7 Kleisli generalization: randomness and nondeterminism

7.1 Monads and Kleisli categories

Let $T = (T, \eta, \mu)$ be a monad on $\mathbf{Set}$ [45, 70]. The Kleisli category $\mathbf{Kl}(T)$ has the same objects as $\mathbf{Set}$ and morphisms $X \to Y$ given by functions $X \to TY$, with Kleisli composition $g \odot f := \mu \circ Tg \circ f$.

Example 7.1 (Nondeterminism: the powerset monad)**.** Let $T = \mathcal{P}$. Then $f : X \to \mathcal{P}(Y)$ is a nondeterministic computation returning a *set* of possible outputs. Kleisli composition is relational composition [45].

Example 7.2 (Randomness: the distribution monad)**.** Let $T = \mathcal{D}$ be the finitary distribution monad on $\mathbf{Set}$: $\mathcal{D}(X)$ is the set of finitely supported probability distributions on X. A Kleisli arrow $f : X \to \mathcal{D}(Y)$ assigns to each input x a distribution over outputs in Y, i.e. a discrete Markov kernel. For measure-theoretic probability, the analogous construction is the Giry monad on measurable spaces [24]; see [30] for the finitary setting and its algebras.

7.2 Cost algebras and Kleisli lifting

To compare objective values under effectful maps, we need a way to *aggregate* a T-structured family of costs. Fix the cost domain $\overline{\mathbb{R}}$ as an object of $\mathbf{Set}$.

Definition 7.3 (Cost algebra)**.** A *cost algebra* for T is a T-algebra $(\overline{\mathbb{R}}, \alpha)$ [70, 44], i.e. a map $\alpha : T\overline{\mathbb{R}} \to \overline{\mathbb{R}}$ satisfying $\alpha \circ \eta_{\overline{\mathbb{R}}} = \mathrm{id}$ and $\alpha \circ T\alpha = \alpha \circ \mu_{\overline{\mathbb{R}}}$.

Definition 7.4 (Kleisli lifting)**.** Given a Kleisli arrow $f : X \to TY$ and a cost function $\varphi : Y \to \overline{\mathbb{R}}$, define

$$f^\dagger(\varphi) := \alpha \circ T\varphi \circ f : X \to \overline{\mathbb{R}}.$$

Lemma 7.5 (Contravariant functoriality of lifting)**.** *For Kleisli arrows $f : X \to TY$ and $g : Y \to TZ$ and any $\psi : Z \to \overline{\mathbb{R}}$,*

$$(g \odot f)^\dagger(\psi) = f^\dagger(g^\dagger(\psi)), \qquad (\eta_X)^\dagger(\varphi) = \varphi.$$

Moreover $f^\dagger$ is monotone with respect to pointwise order on cost functions.

Proof. Using the T-algebra laws,

$$(g \odot f)^\dagger(\psi) = \alpha \circ T\psi \circ \mu \circ Tg \circ f = \alpha \circ T\alpha \circ TT\psi \circ Tg \circ f = \alpha \circ T(\alpha \circ T\psi \circ g) \circ f = f^\dagger(g^\dagger(\psi)).$$

The unit case is $\alpha \circ T\varphi \circ \eta = \varphi$. Monotonicity follows because T and α are monotone on the pointwise order whenever $\overline{\mathbb{R}}$ is viewed as a poset and T is an order-enriched monad; this holds for the standard $\mathcal{P}$ and $\mathcal{D}$ examples. □

Example 7.6 (Standard choices of α)**.** For $T = \mathcal{P}$, common choices are $\alpha = \inf$ ("angelic" nondeterminism: best-case cost) or $\alpha = \sup$ ("demonic": worst-case cost). For $T = \mathcal{D}$, a canonical choice is $\alpha = \mathbb{E}$, expectation of a finitely supported distribution.

7.3 A category of T-effectful optimization reductions

Definition 7.7 (T-reduction morphism)**.** Let $P = (X, c_P)$ and $Q = (Y, c_Q)$. A *T-effectful reduction* $a : P \to Q$ is a pair of Kleisli arrows

$$a = (\mathrm{enc}_a : X \to TY,\ \mathrm{dec}_a : Y \to TX)$$

such that

$$\mathrm{enc}_a^\dagger(c_Q) \le c_P, \tag{4}$$

$$\mathrm{dec}_a^\dagger(c_P) \le c_Q. \tag{5}$$

Proposition 7.8 (Category structure)**.** *T-effectful reductions form a category $\mathbf{OptRed}_{T,\alpha}$ with identities (η_X, η_X) and composition*

$$(b \circ a) := (\mathrm{enc}_b \odot \mathrm{enc}_a,\ \mathrm{dec}_a \odot \mathrm{dec}_b).$$

Proof. Identity laws follow from the unit law of Kleisli composition and Lemma 7.5. For composition, using Lemma 7.5 and monotonicity,

$$(\mathrm{enc}_b \odot \mathrm{enc}_a)^\dagger(c_R) = \mathrm{enc}_a^\dagger(\mathrm{enc}_b^\dagger(c_R)) \le \mathrm{enc}_a^\dagger(c_Q) \le c_P,$$

and similarly

$$(\mathrm{dec}_a \odot \mathrm{dec}_b)^\dagger(c_P) = \mathrm{dec}_b^\dagger(\mathrm{dec}_a^\dagger(c_P)) \le \mathrm{dec}_b^\dagger(c_Q) \le c_R.$$

Associativity is inherited from Kleisli composition. □

7.4 Kleisli-valued functors and natural transformations

Definition 7.9 (Candidate functors into $\mathbf{Kl}(T)$)**.** Define $\mathsf{Cand}_T : \mathbf{OptRed}_{T,\alpha} \to \mathbf{Kl}(T)$ by $\mathsf{Cand}_T(P) = X$ and $\mathsf{Cand}_T(a) = \mathrm{enc}_a$. Define $\mathsf{Cand}_T^{op} : \mathbf{OptRed}_{T,\alpha}^{op} \to \mathbf{Kl}(T)$ by $\mathsf{Cand}_T^{op}(a^{op}) = \mathrm{dec}_a$.

To define feasibility and solution functors into $\mathbf{Kl}(T)$ one needs mild support conditions: a Kleisli arrow $f : X \to TY$ need not map feasible points to feasible points almost surely. We isolate subcategories where this holds.

Definition 7.10 (Support in a subset)**.** Let $f : X \to TY$ and $S \subseteq Y$. We say f *lands in* S if $f(x) \in T(S)$ for all $x \in X$, under the canonical map $T(S) \to T(Y)$ induced by inclusion.

Definition 7.11 (Feasibility-preserving subcategory)**.** Let $\mathbf{OptRed}_{T,\alpha}^{\mathsf{Feas}} \subseteq \mathbf{OptRed}_{T,\alpha}$ be the wide subcategory on the same objects, whose morphisms $a : P \to Q$ satisfy that enc_a lands in $\mathsf{Feas}(Q)$ and dec_a lands in $\mathsf{Feas}(P)$.

Definition 7.12 (Feasible-set functors into $\mathbf{Kl}(T)$)**.** Define $\mathsf{Feas}_T : \mathbf{OptRed}_{T,\alpha}^{\mathsf{Feas}} \to \mathbf{Kl}(T)$ by $\mathsf{Feas}_T(P) = \mathsf{Feas}(P)$ and $\mathsf{Feas}_T(a) = \mathrm{enc}_a|_{\mathsf{Feas}(P)}$ (which is well-defined by Definition 7.10). Similarly define Feas_T^{op} using dec_a.

Definition 7.13 (Optimizer-preserving subcategory)**.** Let $\mathbf{OptRed}_{T,\alpha}^{\mathsf{Sol}} \subseteq \mathbf{OptRed}_{T,\alpha}^{\mathsf{Feas}}$ be the wide subcategory whose morphisms $a : P \to Q$ additionally satisfy that dec_a lands in $\mathsf{Sol}(P)$ when restricted to $\mathsf{Sol}(Q)$ (in the sense of Definition 7.10).

Definition 7.14 (Solution-set functors into $\mathbf{Kl}(T)$)**.** Define $\mathsf{Sol}_T : (\mathbf{OptRed}_{T,\alpha}^{\mathsf{Sol}})^{op} \to \mathbf{Kl}(T)$ by $\mathsf{Sol}_T(P) = \mathsf{Sol}(P)$ and $\mathsf{Sol}_T(a^{op}) = \mathrm{dec}_a|_{\mathsf{Sol}(Q)}$.

7.5 Naturality in the Kleisli world

In $\mathbf{Kl}(T)$, a "natural transformation" has components that are themselves Kleisli arrows.

Definition 7.15 (Kleisli-natural transformation)**.** Let $F, G : \mathcal{C} \to \mathbf{Kl}(T)$ be functors. A *Kleisli-natural transformation* $\tau : F \Rightarrow G$ is a family of Kleisli arrows $\tau_C : F(C) \to T(G(C))$ such that for every $f : C \to D$ in $\mathcal{C}$,

$$G(f) \odot \tau_C \ = \ \tau_D \odot F(f)$$

in $\mathbf{Kl}(T)$.

Theorem 7.16 (Inclusions in $\mathbf{Kl}(T)$)**.** *On* $\mathbf{OptRed}^{\mathsf{Feas}}_{T,\alpha}$ *there is a Kleisli-natural transformation* $\iota^T : \mathsf{Feas}_T \Rightarrow \mathsf{Cand}_T$ *with components* $\iota^T_P := \eta_{\mathsf{Cand}(P)} \circ \mathrm{inc} : \mathsf{Feas}(P) \to T(\mathsf{Cand}(P))$ *induced by set inclusion. Similarly, on* $\mathbf{OptRed}^{\mathsf{Sol}}_{T,\alpha}$ *there is* $\kappa^T : \mathsf{Sol}_T \Rightarrow \mathsf{Feas}^{op}_T$ *induced by inclusion.*

Proof. Since ι^T_P is pure (via η), the Kleisli naturality condition reduces to the commutativity of the underlying set-theoretic squares, as in Theorem 5.1. The same reasoning applies to κ^T. □

7.6 Costs as oplax naturality via Kleisli lifting

Let $\mathcal{E}_T : \mathbf{Kl}(T)^{op} \to \mathbf{Pos}$ map X to $(\overline{\mathbb{R}})^X$ and a Kleisli arrow $f : X \to TY$ to the Kleisli lifting $f^\dagger : (\overline{\mathbb{R}})^Y \to (\overline{\mathbb{R}})^X$. Lemma 7.5 states that $\mathcal{E}_T$ is a functor. Then (4)–(5) can be read as oplax naturality conditions in $\mathbf{Pos}$, exactly as in Proposition 6.2.

8 Approximate reductions

Exact reductions are often too rigid: typical algorithms guarantee solutions whose objective value is within a multiplicative and/or additive slack of the target value. The categorical bookkeeping below is closely related to approximation-preserving reductions in complexity theory [71, 58, 57, 22].

8.1 Affine slack parameters as a monoid

Throughout this section we restrict to nonnegative costs $\overline{\mathbb{R}}_{\geq 0} := [0, \infty] = \mathbb{R}_{\geq 0} \cup \{+\infty\}$. (If costs may be negative, multiplicative approximation factors are usually ill-behaved.)

Definition 8.1 (Affine-approximation monoid)**.** Let $\mathbf{Aff}_{\geq 0}$ be the monoid of pairs (α, β) with $\alpha \geq 0$ and $\beta \geq 0$, with multiplication

$$(\alpha, \beta) \otimes (\alpha', \beta') := (\alpha\alpha',\ \beta + \alpha\beta'),$$

and unit $e := (1, 0)$. Identify (α, β) with the monotone affine map $f_{\alpha,\beta} : \overline{\mathbb{R}}_{\geq 0} \to \overline{\mathbb{R}}_{\geq 0}$, $t \mapsto \alpha t + \beta$.

Lemma 8.2 (Systematic composition)**.** *For all* $(\alpha, \beta), (\alpha', \beta') \in \mathbf{Aff}_{\geq 0}$,

$$f_{\alpha,\beta} \circ f_{\alpha',\beta'} = f_{(\alpha,\beta)\otimes(\alpha',\beta')}.$$

Hence multiplicative factors multiply and additive terms accumulate (with the expected scaling by upstream factors).

Proof. $f_{\alpha,\beta}(f_{\alpha',\beta'}(t)) = \alpha(\alpha' t + \beta') + \beta = (\alpha\alpha')t + (\beta + \alpha\beta')$. □

8.2 Deterministic approximate reductions

Definition 8.3 (Approximate reduction morphism)**.** Let $P = (X, c_P)$ and $Q = (Y, c_Q)$ with $c_P, c_Q : X, Y \to \overline{\mathbb{R}}_{\geq 0}$. An *approximate reduction morphism* $a : P \to Q$ is a triple

$$a = (\mathrm{enc}_a : X \to Y,\ \mathrm{dec}_a : Y \to X,\ \gamma_a \in \mathbf{Aff}_{\geq 0})$$

such that

$$c_Q(\mathrm{enc}_a(x)) \leq c_P(x) \quad \text{for all } x \in X, \tag{6}$$

$$c_P(\mathrm{dec}_a(y)) \leq f_{\gamma_a}(c_Q(y)) = \alpha_a c_Q(y) + \beta_a \quad \text{for all } y \in Y, \tag{7}$$

where $\gamma_a = (\alpha_a, \beta_a)$.

Proposition 8.4 (Composition of approximate bounds)**.** *If* $a : P \to Q$ *and* $b : Q \to R$ *are approximate reductions with slack parameters* $\gamma_a = (\alpha_a, \beta_a)$ *and* $\gamma_b = (\alpha_b, \beta_b)$*, then the composite* $b \circ a : P \to R$ *satisfies*

$$c_P(\mathrm{dec}_{b\circ a}(z)) \leq (\alpha_a\alpha_b)\, c_R(z) + (\beta_a + \alpha_a\beta_b),$$

i.e. $\gamma_{b\circ a} = \gamma_a \otimes \gamma_b$.

Proof. Using $\mathrm{dec}_{b\circ a} = \mathrm{dec}_a \circ \mathrm{dec}_b$,

$$c_P(\mathrm{dec}_a(\mathrm{dec}_b(z))) \leq \alpha_a c_Q(\mathrm{dec}_b(z)) + \beta_a \leq \alpha_a(\alpha_b c_R(z) + \beta_b) + \beta_a = (\alpha_a\alpha_b)c_R(z) + (\beta_a + \alpha_a\beta_b).$$

□

Proposition 8.5 (Category and grading)**.** *Approximate reductions form a category* $\mathbf{OptRed}_{\approx}$ *with the same objects as* $\mathbf{OptRed}_{\mathrm{ex}}$ *and morphisms as in Definition 8.3. Composition is*

$$(b \circ a) := (\mathrm{enc}_b \circ \mathrm{enc}_a,\ \mathrm{dec}_a \circ \mathrm{dec}_b,\ \gamma_a \otimes \gamma_b),$$

and identities are $(\mathrm{id}_X, \mathrm{id}_X, (1, 0))$.

Moreover, the assignment $a \mapsto \gamma_a$ *is functorial from the* opposite *category:*

$$\Gamma : \mathbf{OptRed}_{\approx}^{op} \to \mathbf{BAff}_{\geq 0}, \qquad \Gamma(a^{op}) = \gamma_a,$$

where $\mathbf{BAff}_{\geq 0}$ *is the one-object category associated to the monoid* $\mathbf{Aff}_{\geq 0}$.

Proof. Associativity follows from associativity of function composition and of $\otimes$ (Lemma 8.2). For the grading, note $\Gamma((b \circ a)^{op}) = \gamma_{b \circ a} = \gamma_a \otimes \gamma_b = \Gamma(a^{op}) \circ \Gamma(b^{op})$ in $\mathbf{BAff}_{\geq 0}$. □

8.3 Approximation in the Kleisli world

In the effectful setting, (7) is interpreted via Kleisli lifting: for $f : Y \to TX$, $f^\dagger(c_P)$ is the *aggregated* cost on Y induced by f.

To propagate affine bounds through Kleisli liftings, it suffices that affine maps are homomorphisms of the chosen cost algebra.

Definition 8.6 (Affine-homomorphic cost algebra)**.** Let $(\overline{\mathbb{R}}_{\geq 0}, \alpha)$ be a T-algebra (Section 7.2, with $\overline{\mathbb{R}}$ replaced by $\overline{\mathbb{R}}_{\geq 0}$). We say it is *affine-homomorphic* if for every $(\alpha', \beta') \in \mathbf{Aff}_{\geq 0}$, the affine map $f_{\alpha',\beta'} : \overline{\mathbb{R}}_{\geq 0} \to \overline{\mathbb{R}}_{\geq 0}$ is a T-algebra homomorphism, i.e.

$$\alpha \circ T(f_{\alpha',\beta'}) = f_{\alpha',\beta'} \circ \alpha.$$

This holds for $(T, \alpha) = (\mathcal{P}, \inf)$, $(\mathcal{P}, \sup)$, and $(\mathcal{D}, \mathbb{E})$ (Example 7.6).

Lemma 8.7 (Affine maps commute with Kleisli lifting)**.** *Assume* $(\overline{\mathbb{R}}_{\geq 0}, \alpha)$ *is affine-homomorphic and let* $f : X \to TY$ *be a Kleisli arrow. Then for any* $\varphi : Y \to \overline{\mathbb{R}}_{\geq 0}$ *and any* $(\alpha', \beta') \in \mathbf{Aff}_{\geq 0}$,

$$f^\dagger(f_{\alpha',\beta'} \circ \varphi) = f_{\alpha',\beta'} \circ f^\dagger(\varphi).$$

Proof.

$$f^\dagger(f_{\alpha',\beta'} \circ \varphi) = \alpha \circ T(f_{\alpha',\beta'} \circ \varphi) \circ f = \alpha \circ T(f_{\alpha',\beta'}) \circ T\varphi \circ f = f_{\alpha',\beta'} \circ \alpha \circ T\varphi \circ f = f_{\alpha',\beta'} \circ f^\dagger(\varphi).$$

□

Definition 8.8 (T-approximate reduction)**.** Let $P = (X, c_P)$ and $Q = (Y, c_Q)$ with costs into $\overline{\mathbb{R}}_{\geq 0}$, and let $(\overline{\mathbb{R}}_{\geq 0}, \alpha)$ be an affine-homomorphic T-algebra. A *T-approximate reduction* $a : P \to Q$ is data

$$a = (\mathrm{enc}_a : X \to TY,\ \mathrm{dec}_a : Y \to TX,\ \gamma_a \in \mathbf{Aff}_{\geq 0})$$

such that $\mathrm{enc}_a^\dagger(c_Q) \leq c_P$ and

$$\mathrm{dec}_a^\dagger(c_P) \leq f_{\gamma_a} \circ c_Q.$$

Proposition 8.9 (Composition of T-approximate reductions)**.** *Under the assumptions of Definition 8.8,* T*-approximate reductions are closed under composition in the sense that if* $a : P \to Q$ *and* $b : Q \to R$ *are* T*-approximate reductions then* $b \circ a : P \to R$ *is a* T*-approximate reduction with* $\gamma_{b \circ a} = \gamma_a \otimes \gamma_b$.

Proof. The encoding inequality is as in Proposition 7.8. For decoding, using $\mathrm{dec}_{b \circ a} = \mathrm{dec}_a \odot \mathrm{dec}_b$ and Lemma 7.5,

$$(\mathrm{dec}_a \odot \mathrm{dec}_b)^\dagger(c_P) = \mathrm{dec}_b^\dagger(\mathrm{dec}_a^\dagger(c_P)) \leq \mathrm{dec}_b^\dagger(f_{\gamma_a} \circ c_Q) = f_{\gamma_a} \circ \mathrm{dec}_b^\dagger(c_Q) \leq f_{\gamma_a} \circ f_{\gamma_b} \circ c_R = f_{\gamma_a \otimes \gamma_b} \circ c_R,$$

where we used Lemma 8.7 in the equality and Lemma 8.2 in the last step. □

9 Resources and enrichment

Besides correctness/quality guarantees, optimization pipelines are governed by *resources* such as runtime and memory. Categorically, resources are naturally tracked by monoidal or enriched structure [44, 33, 17].

9.1 A resource monoidal preorder

Definition 9.1 (Resource domain)**.** A *resource domain* is a commutative monoidal preorder $(W, \leq, \otimes, I)$. A basic example for (time, memory) is

$$W := (\overline{\mathbb{R}}_{\geq 0})^2, \qquad (t, m) \otimes (t', m') := (t + t',\, m + m'), \qquad I = (0, 0),$$

with componentwise order.

We write $\mathbf{B}W$ for the one-object category (delooping) of the monoid $(W, \otimes, I)$.

9.2 Resource-annotated reductions

Definition 9.2 (Resource-annotated reduction category)**.** Fix a category of reductions $\mathcal{C}$ (e.g. $\mathbf{OptRed}_{\mathrm{ex}}$, $\mathbf{OptRed}_{T,\alpha}$, or $\mathbf{OptRed}_{\approx}$) and a resource domain $(W, \leq, \otimes, I)$. Define $\mathcal{C}_W$ to have:

- the same objects as $\mathcal{C}$,
- morphisms $P \to Q$ given by pairs (a, w) with $a \in \mathcal{C}(P, Q)$ and $w \in W$,
- identity $\mathrm{id}_P = (\mathrm{id}_P, I)$,
- composition $(b, w_b) \circ (a, w_a) := (b \circ a,\ w_a \otimes w_b)$.

When $\mathcal{C} = \mathbf{OptRed}_{\mathrm{ex}}$ we write $\mathbf{OptRed}_{\mathrm{res}} := \mathcal{C}_W$ and simply denote it by $\mathbf{OptRed}_{\mathrm{res}}$.

Proposition 9.3 (Bookkeeping functors)**.** *There are functors*

$$U : \mathbf{OptRed}_{\mathrm{res}} \to \mathbf{OptRed}_{\mathrm{ex}}, \qquad U(a, w) = a,$$

and

$$\mathsf{Res} : \mathbf{OptRed}_{\mathrm{res}} \to \mathbf{B}W, \qquad \mathsf{Res}(a, w) = w,$$

with $\mathsf{Res}((b, w_b) \circ (a, w_a)) = \mathsf{Res}(a, w_a) \otimes \mathsf{Res}(b, w_b)$. *Thus resource usage of a composed pipeline is the monoidal product (e.g. sum) of its components.*

Proof. Functoriality of U is immediate. For Res, note $\mathsf{Res}(\mathrm{id}_P) = I$ and $\mathsf{Res}((b, w_b) \circ (a, w_a)) = w_a \otimes w_b = \mathsf{Res}(a, w_a) \circ \mathsf{Res}(b, w_b)$ in the one-object category $\mathbf{B}W$. □

Remark 9.4 (Reduce–solve–recover)**.** If one models a pipeline as a composite

$$P \xrightarrow{(\text{reduce}, w_r)} Q \xrightarrow{(\text{solve}, w_s)} Q \xrightarrow{(\text{recover}, w_c)} P,$$

then Proposition 9.3 yields the total resource bound $w_r \otimes w_s \otimes w_c$, without additional bookkeeping. This is the categorical version of "costs add along the pipeline".

9.3 Enrichment viewpoint: minimal resource as a Lawvere metric

Given $(\mathbf{OptRed}_{\mathrm{res}}, \mathsf{Res})$ one can extract *minimal* resource requirements between problems by taking an infimum.

Proposition 9.5 (Resource enrichment)**.** *Let $(W, \leq, \otimes, I)$ be a commutative monoidal preorder where infima exist. Define for $P, Q \in \mathbf{OptRed}_{\mathrm{ex}}$:*

$$d(P, Q) := \inf\{\, w \in W : \exists\, (a, w) \in \mathbf{OptRed}_{\mathrm{res}}(P, Q) \,\}.$$

Then $d(P, P) \leq I$ and $d(P, R) \leq d(P, Q) \otimes d(Q, R)$, so $\mathbf{OptRed}_{\mathrm{ex}}$ becomes a W-enriched category in the sense of Lawvere [37] (see also [33]).

Proof. $d(P, P) \leq I$ because $(\mathrm{id}_P, I) \in \mathbf{OptRed}_{\mathrm{res}}(P, P)$. For the triangle inequality, given $(a, w_a) : P \to Q$ and $(b, w_b) : Q \to R$, their composite is $(b \circ a, w_a \otimes w_b) : P \to R$, so taking infima yields $d(P, R) \leq d(P, Q) \otimes d(Q, R)$. □

9.4 Internalizing resources via a writer monad and transformer

An alternative to external resource annotations is to *internalize* resource accounting into the Kleisli world. Given a commutative monoid $(W, \otimes, I)$, the *writer monad* $W \times (-)$ on $\mathbf{Set}$ sends X to $W \times X$ and accumulates resources by $\otimes$ under Kleisli composition [45, 72].

More generally, to combine resources with an ambient effect monad T (e.g. nondeterminism or randomness), one uses the *writer monad transformer* (in Haskell notation, `WriterT`)

$$T^W(X) := T(X \times W), \tag{8}$$

whose Kleisli arrows $X \to T(Y \times W)$ compute an output in Y and a resource log in W. Kleisli composition multiplies the logs using $\otimes$ [40, 72]. Section 10 below exploits the same idea with a *product* log that simultaneously tracks approximation loss and runtime/memory.

10 A unified graded/effectful structure

This section combines the three extensions above into a single compositional bookkeeping device: *effects* via a monad T (Section 7), *approximation slack* via the affine monoid $\mathbf{Aff}_{\geq 0}$ (Section 8), and *resources* via a monoidal preorder $(W, \otimes, I)$ (Section 9). The result is a "one-line" composition rule that simultaneously tracks quality loss and runtime/memory. Conceptually, this is a very simple instance of the general graded/parameterised monad viewpoint used in effect systems [2, 32, 55].

10.1 The product grade monoid $\mathbf{Aff}_{\geq 0} \times W$

Definition 10.1 (Product grade)**.** Let $(W, \leq, \otimes, I)$ be a commutative monoidal preorder and let $\mathbf{Aff}_{\geq 0}$ be the affine approximation monoid (Definition 8.1). Define the *grade monoid*

$$G := \mathbf{Aff}_{\geq 0} \times W$$

with multiplication and unit given componentwise:

$$(\gamma, w) \otimes_G (\gamma', w') := (\gamma \otimes \gamma',\ w \otimes w'), \qquad e_G := (e, I),$$

where $e = (1, 0)$ is the unit of $\mathbf{Aff}_{\geq 0}$. When we need an order on G, we use the product order induced by $(\mathbf{Aff}_{\geq 0}, \leq)$ (componentwise on (α, β)) and $(W, \leq)$.

Remark 10.2. Writing $\gamma = (\alpha, \beta)$ and $w = (t, m)$ (time/memory), the product multiplication is exactly the desired "single rule":

$$(\alpha, \beta, t, m) \otimes_G (\alpha', \beta', t', m') = (\alpha\alpha',\ \beta + \alpha\beta',\ t + t',\ m + m').$$

Multiplicative approximation factors multiply; additive error terms accumulate (with the upstream scaling); and resources add.

10.2 Externally graded effectful reductions

Fix a monad T on $\mathbf{Set}$ and an affine-homomorphic cost algebra $(\overline{\mathbb{R}}_{\geq 0}, \alpha)$ as in Definition 8.6.

Definition 10.3 (Graded, effectful, approximate, resource-aware reduction)**.** Let $P = (X, c_P)$ and $Q = (Y, c_Q)$ with $c_P, c_Q : X, Y \to \overline{\mathbb{R}}_{\geq 0}$. A *graded T-reduction*

$$a : P \to Q$$

is a tuple

$$a = (\mathrm{enc}_a : X \to TY,\ \mathrm{dec}_a : Y \to TX,\ g_a = (\gamma_a, w_a) \in G)$$

such that

$$\mathrm{enc}_a^\dagger(c_Q) \leq c_P, \tag{9}$$

$$\mathrm{dec}_a^\dagger(c_P) \leq f_{\gamma_a} \circ c_Q. \tag{10}$$

Intuitively, γ_a is an *a priori* approximation guarantee for recovery and w_a is an *a priori* resource bound.

Proposition 10.4 (Category structure and a single grade law)**.** *Graded T-reductions form a category $\mathbf{OptRed}^{\mathrm{gr}}_{T,\alpha}$ with the same objects as $\mathbf{OptRed}_{T,\alpha}$ and:*

- *identity on $P = (X, c_P)$: (η_X, η_X, e_G),*

- *composition: for $a : P \to Q$ and $b : Q \to R$,*

$$(b \circ a) := (\mathrm{enc}_b \odot \mathrm{enc}_a,\ \mathrm{dec}_a \odot \mathrm{dec}_b,\ g_a \otimes_G g_b).$$

In particular, the grade of a composite pipeline is the monoid product of the grades of its steps.

Proof. Closure of the encoding inequality is as in Proposition 7.8. For the decoding inequality, Proposition 8.9 shows that approximation parameters compose by $\gamma_{b\circ a} = \gamma_a \otimes \gamma_b$, and resource parameters compose by $w_{b\circ a} = w_a \otimes w_b$ by definition. Associativity follows from associativity of Kleisli composition and of $\otimes_G$. □

Proposition 10.5 (Grading/measurement functor)**.** *There is a functor*

$$\mathsf{Grade} : \mathbf{OptRed}^{\mathrm{gr}}_{T,\alpha}{}^{op} \to \mathbf{B}G, \qquad \mathsf{Grade}(a^{op}) := g_a,$$

where $\mathbf{B}G$ is the one-object category of the monoid G.

Proof. By Proposition 10.4, $\mathsf{Grade}((b \circ a)^{op}) = g_{b\circ a} = g_a \otimes_G g_b$, which is composition in $\mathbf{B}G$. □

Remark 10.6 (Recovering earlier frameworks)**.** Forgetting parts of the grade gives canonical functors:

- $\mathbf{OptRed}^{\mathrm{gr}}_{T,\alpha} \to \mathbf{OptRed}_{T,\alpha}$ (forget g_a),
- $\mathbf{OptRed}^{\mathrm{gr}}_{T,\alpha} \to$ "T-approximate reductions" (forget w_a),
- $\mathbf{OptRed}^{\mathrm{gr}}_{T,\alpha} \to$ "resource-annotated effectful reductions" (forget γ_a).

Thus $\mathbf{OptRed}^{\mathrm{gr}}_{T,\alpha}$ packages all three bookkeeping layers into a single compositional object.

10.3 Internalizing the grade via a writer transformer

Instead of annotating arrows by a grade externally, one can internalize the grade into the ambient effect monad itself. Let $G = \mathbf{Aff}_{\geq 0} \times W$ be the grade monoid.

Definition 10.7 (Writer transformer for a grade monoid)**.** Given a monad T on $\mathbf{Set}$ and a commutative monoid $(G, \otimes_G, e_G)$, define the *writer transformer* endofunctor

$$T^G(X) := T(X \times G)$$

as in (8) (with W replaced by G). Its Kleisli arrows $X \to T(Y \times G)$ return both an output and a grade log. Kleisli composition multiplies logs with $\otimes_G$ [40, 72].

Remark 10.8 (How it matches external grading)**.** If a morphism in $\mathbf{OptRed}^{\mathrm{gr}}_{T,\alpha}$ has a *fixed* grade $g_a \in G$ (independent of input), it can be encoded as a Kleisli arrow for T^G by "tagging" each output with g_a:

$$\widetilde{\mathrm{enc}}_a := T(\lambda y.(y, g_a)) \circ \mathrm{enc}_a : X \to T(Y \times G), \qquad \widetilde{\mathrm{dec}}_a := T(\lambda x.(x, g_a)) \circ \mathrm{dec}_a : Y \to T(X \times G).$$

Then the writer-transformer composition law ensures that the grade tags combine automatically as $g_{b\circ a} = g_a \otimes_G g_b$, reproducing Proposition 10.4 inside the Kleisli category $\mathbf{Kl}(T^G)$.

Remark 10.9 (Stochastic or nondeterministic grades)**.** The same internalization naturally supports variable grades: for nondeterminism, $f : X \to \mathcal{P}(Y \times G)$ can return multiple possible (output,grade) pairs; for randomness, $f : X \to \mathcal{D}(Y \times G)$ returns a distribution over them. If desired, one may aggregate such variable grades (e.g. worst-case or expectation) using an additional algebra on G; this is a lightweight instance of graded/parameterised effects [2, 32, 55].

11 Application instantiations

The constructions above are intentionally *representation-agnostic*: an object is a costed space $P = (X, c_P)$ and a morphism is a certified "reduce–solve–recover" procedure, possibly effectful (Kleisli), approximate (affine slack), and resource-aware (graded). For concrete work one typically restricts to *full subcategories* whose objects carry additional structure (vector spaces, convexity, smoothness, oracle models, populations, ...) and whose morphisms are required to respect that structure. This section sketches several application-driven instantiations; the point is not to be exhaustive but to show that a single categorical interface can cover classical deterministic methods, stochastic methods, and modern heuristics.

11.1 Convex optimization as a structured subcategory

Let $(\mathbf{OptRed}^{\mathrm{gr}}_{T,\alpha})_{\mathrm{cvx}}$ denote the full subcategory of $\mathbf{OptRed}^{\mathrm{gr}}_{T,\alpha}$ whose objects are convex optimization problems in the usual analytic sense: $X \subseteq \mathbb{R}^n$ is convex and $c_P : X \to \overline{\mathbb{R}}$ is proper, lower semicontinuous, and convex (with constraints again encoded by $+\infty$) [64, 9, 49]. Morphisms in $(\mathbf{OptRed}^{\mathrm{gr}}_{T,\alpha})_{\mathrm{cvx}}$ are those graded/effectful reductions whose encoding/decoding maps preserve the convex structure; typical examples include:

- affine changes of variables and introduction/elimination of slack variables (exact reductions);
- epigraph transformations, introducing an auxiliary variable t with constraint $c(x) \leq t$, which is the canonical way to express composite objectives and leads to conic formulations [9];
- reductions to standard cones (LP/SOCP/SDP) and back, which can be treated functorially as "compilers" from a high-level convex modeling language to a solver input format.

Within such a subcategory one can also treat *duality* as a (partially defined) contravariant endofunctor on the subcategory where a suitable regularity condition holds (e.g. Slater-type conditions), with a comparison map $P \to D(D(P))$ expressing "bidual recovery" [64, 50]. First-order and interior-point methods can be modeled as (resource-graded) endomorphisms $P \to P$ in appropriate state spaces, and their equivalences (e.g. algebraically identical updates written in different coordinates) are naturally treated as higher morphisms in the later (∞, n) framework.

11.2 Nonconvex optimization and relax–round pipelines

Let $(\mathbf{OptRed}^{\mathrm{gr}}_{T,\alpha})_{\mathrm{nc}}$ denote the full subcategory where $X \subseteq \mathbb{R}^n$ and c_P is arbitrary (possibly nonconvex and nonsmooth). The characteristic feature of nonconvex optimization is that global solution recovery is rarely exact; instead one uses *relaxations* (often convex) that provide bounds, together with a decoding/rounding procedure that returns a feasible point for the original problem. Categorically, this is precisely a morphism

$$a : P \to Q$$

in the *approximate* sense of Section 8, where Q is a tractable relaxation and the grade $\gamma_a = (\alpha, \beta)$ quantifies the loss incurred when decoding a relaxed solution back to the original space. This covers classical "relax–solve–round" patterns in approximation algorithms [71, 58] and also modern convex-relaxation heuristics.

Local methods (gradient descent, trust-region methods, coordinate descent) fit naturally into the *effectful* picture: random initialization and stochastic perturbations become Kleisli arrows $X \to \mathcal{D}(X)$, while multistart or branching strategies become $X \to \mathcal{P}(X)$ [53, 7]. The resource component of the grade can track iteration budgets or function/gradient evaluations, and the approximation component can encode a target stationarity notion (e.g. $\|\nabla f(x)\|$ small) as a surrogate objective in the extended-real encoding.

11.3 Linear programming and the simplex family

Linear programs form an important application domain where reduction and representation change are central. Let LP be a category whose objects are LP instances (equivalently, polyhedra with linear costs) and whose morphisms are the standard exact preprocessing transformations (scaling, redundancy elimination, conversion between inequality and equality form via slacks, etc.) [12]. The simplex method itself is most naturally modeled not as a single arrow but as a *dynamical system on representations*: for a fixed pivot rule, a pivot step is a (partial) function on tableaux/bases; for an *nondeterministic* pivot choice, it is a Kleisli arrow

$$\mathsf{Tableau} \to \mathcal{P}(\mathsf{Tableau})$$

returning the set of admissible next pivots. Composing such arrows corresponds to multiple pivot steps, and a resource grade can count pivots or arithmetic operations. Distinct pivot rules (Bland, steepest-edge, ...) can be modeled as different endomorphisms; 2-cells or higher cells can witness that two implementations realize the same pivot map up to certified tableau isomorphism. The Klee–Minty construction [34] can be read here as an "adversarial object" showing that certain resource grades can grow exponentially in the worst case.

11.4 Stochastic optimization and stochastic programming

Many modern problems take the form

$$\min_{x \in X} \mathbb{E}[F(x, \xi)],$$

or more generally stochastic programs with recourse [67]. The Kleisli semantics for the distribution monad is canonical here: an algorithmic update that maps a current iterate to a random next iterate is a Markov kernel $K : X \to \mathcal{D}(X)$, and k iterations are just the Kleisli composite $K^{\odot k}$ [24, 30]. Classical stochastic approximation [62, 35] and modern large-scale methods such as SGD and its variants [8] can be expressed in this form once the state space is enriched to include step-size schedules, momentum variables, or variance-reduction state.

The unified grade $G = \mathbf{Aff}_{\geq 0} \times W$ is useful operationally: the resource component W can track sample complexity or number of gradient evaluations, while the affine component can encode an a priori bound on expected suboptimality or a high-probability error target. In this view, "variance reduction" and "mini-batching" are morphisms that trade resource grade against approximation grade in a compositional way.

11.5 Heuristic and evolutionary optimization

Heuristic methods (simulated annealing, tabu search, genetic and evolutionary algorithms) are often used when exact certificates are unavailable or too expensive. In our framework they still fit cleanly as *effectful, graded* morphisms: a population-based method operates on a state space of populations (e.g. X^N or finite multisets), and one iteration is naturally a Markov kernel

$$\mathsf{Pop}(X) \to \mathcal{D}(\mathsf{Pop}(X))$$

built by composing kernels for selection, crossover, and mutation [28, 25, 15]. The resource grade tracks the dominant cost (fitness evaluations), while the approximation grade can be interpreted in several ways: as a weak guarantee when available (e.g. expected improvement in a surrogate cost), or simply as a *measured* or *empirical* quality annotation attached to a pipeline. The higher-categorical layer is again useful: many implementations differ only by algebraic rebracketing of operators (e.g. applying mutation before crossover) or by semantics-preserving compiler rewrites; these become higher equivalences once a notion of weak equivalence W is chosen (Sections 16 and 17).

Summary. Across these domains, the categorical viewpoint provides a uniform language: preprocessing and reformulation are morphisms, solver iterations are endomorphisms in an effectful world, approximation and runtime/memory are tracked by a single grade, and implementation rewrites live as higher morphisms. The benefit is compositional bookkeeping: once each component is certified/graded, whole pipelines inherit guarantees by categorical composition.

12 Optimization transformations for ensemble and averaging methods in machine learning

Ensemble and averaging methods are often presented as "model-combination" heuristics, but many of them are naturally instances of the reduce–solve–recover pattern and can be expressed directly as optimization problems in the present categorical semantics. The key move is to treat an ensemble either as (i) a *distribution* over predictors (a $\mathcal{D}$-object), or (ii) a *finite convex combination* of a library of predictors (a point on a simplex). Learning (or choosing) the ensemble then becomes an optimization problem over these combination variables, while stochastic resampling and random initialization appear as Kleisli effects.

12.1 Supervised learning objects and ensemble decision spaces

Fix a supervised dataset $D = \{(x_i, y_i)\}_{i=1}^n$, a hypothesis/predictor class $\mathcal{H}$, a loss ℓ, and a regularizer R. The basic empirical-risk problem is an object

$$P_D = (\mathcal{H}, c_D), \qquad c_D(h) := \frac{1}{n}\sum_{i=1}^{n} \ell(h(x_i), y_i) + \lambda R(h),$$

which is a standard instance of the extended-real encoding (constraints are absorbed by $+\infty$) [9, 64].

Finite weighted averaging. Given a *library* of base predictors $h_1, \ldots, h_m \in \mathcal{H}$, a common ensemble class is the simplex of weight vectors

$$\Delta_m := \Big\{ w \in \mathbb{R}^m_{\geq 0} : \sum_{j=1}^{m} w_j = 1 \Big\},$$

with combined predictor (for regression, or for class-probability models)

$$h_w(x) := \sum_{j=1}^{m} w_j \, h_j(x).$$

Choosing weights by validation or cross-validation is then the *meta-optimization* problem

$$Q_{D,h_{1:m}} := (\Delta_m, c_{\text{ens}}), \qquad c_{\text{ens}}(w) := \frac{1}{n}\sum_{i=1}^{n} \ell(h_w(x_i), y_i) + \lambda_w \Omega(w).$$

When ℓ is convex in its first argument and $h_w(x)$ is affine in w, the map $w \mapsto c_{\text{ens}}(w)$ is convex, so the ensemble-weight step is itself a convex optimization problem (often a small one) that composes cleanly with upstream training [9]. This encompasses linear stacking/metalearning, where the features are base-model predictions and the meta-learner solves an ERM over the simplex or an unconstrained weight space [73, 13].

Distributional ensembles. More abstractly, an ensemble can be represented as a (finite-support) distribution $\pi \in \mathcal{D}(\mathcal{H})$ over base predictors, with prediction interpreted as a mixture. This moves ensembling into the same effectful semantics used elsewhere: the "decision" is itself a distribution, and expected loss is computed using the $\mathcal{D}$-algebra $\alpha = \mathbb{E}$ on $\overline{\mathbb{R}}$ [24, 30]. Bayesian model averaging is the canonical example in which π is a posterior over models/parameters and prediction is posterior averaging [27].

12.2 Bagging, random forests, and bootstrap aggregation as Kleisli morphisms

Bagging (bootstrap aggregation) trains base predictors on bootstrap-resampled datasets and averages them [10]. Categorically, the bootstrap step is an *effectful encoding* of data:

$$D \longmapsto \mathcal{D}(\mathcal{D}), \qquad D' \sim \text{Bootstrap}(D),$$

and "train a model on a dataset" is itself a (typically approximate) solve step producing $h \in \mathcal{H}$. The overall pipeline can be viewed as a Kleisli arrow

$$D \;\longrightarrow\; \mathcal{D}(\mathcal{H}^m)$$

returning a distribution over m-tuples of trained predictors (one per bootstrap draw), followed by a deterministic recovery map $\mathcal{H}^m \to \mathcal{H}_{\text{avg}}$ that averages predictions. In the present framework this is exactly a reduce–solve–recover composite in $\mathbf{Kl}(()\mathcal{D})$: the randomness lives in the encoding stage, while the decoding stage is an averaging functional.

Random forests instantiate the same pattern with additional internal randomness (feature subsampling and randomized splits) and a particular base learner (decision trees) [11]. The graded structure is natural here: the resource component records training cost for m base learners and inference cost for evaluating all base models, while the approximation component can encode bounds relative to the best convex combination in the chosen library.

12.3 Boosting as optimization over convex hulls and stagewise composition

Boosting constructs an additive (often sparse) ensemble by iteratively selecting and reweighting weak learners; AdaBoost is a classical example [18]. A useful view is that the *target* object is an optimization problem over the convex hull (or linear span) of a weak-learner class, and boosting is a *stagewise* approximate solver for it. For example, in many settings one chooses coefficients α_t and base predictors h_t so that the combined predictor is

$$H_T(x) := \sum_{t=1}^{T} \alpha_t \, h_t(x),$$

and the coefficients/learners are chosen to decrease a convex surrogate loss. Gradient boosting makes this explicit by interpreting the procedure as functional gradient descent on a risk functional [20].

In our categorified language, each boosting round is a resource-graded endomorphism on an appropriate state space that stores the current ensemble and the current reweighting of data; composing T rounds is categorical composition. The approximation grade records the suboptimality of the stagewise procedure relative to the optimum in the span/hull, and the resource grade records the number of rounds and base-learner training cost.

12.4 Deep ensembles, checkpoint averaging, and weight-space averaging

In deep learning, a widely used uncertainty baseline is to train multiple networks from different random initializations and average their predictive distributions [36]. This is a direct instance of the distributional-ensemble semantics above, with the randomness implemented as a Kleisli effect in training. Variants such as snapshot ensembles or stochastic weight averaging (SWA) replace an explicit ensemble by averaging parameters along a training trajectory; empirically this often improves generalization [29]. From the present viewpoint, SWA is a deterministic "recovery" map that projects a path (or multiset) of parameter iterates to a single parameter vector, while its interpretation as approximating a posterior or an averaged predictor is naturally expressed in the $\mathcal{D}$-based semantics.

Summary. Ensembling is therefore not outside the optimization categorification; it is a particularly clear instance where *changing the decision space* (from a single predictor to a convex combination or distribution over predictors) turns a model-combination rule into a well-formed optimization problem. Randomized resampling and randomized training fit the Kleisli effect layer, weight learning fits the convex subcategory, and the unified grade $G = \mathbf{Aff}_{\geq 0} \times W$ bookkeeps the quality–resource trade-offs of large ensembles.

13 Categorifying game theory and integrating optimization reductions

Game theory is, at its core, the study of *interacting* optimization problems: each player optimizes an objective that depends on the choices of the others. This section develops a categorical semantics for games that is compatible with the optimization-reduction categories built above, and then exhibits functorial bridges between the two worlds. We focus on noncooperative games (normal-form and several standard extensions), because their solution concepts (Nash, correlated, stochastic equilibria) are the ones most directly connected to optimization and variational inequalities [48, 47, 56, 16].

13.1 Normal-form games in cost form

To align game theory with the minimization-centric optimization categories in Sections 2–9, we work with *cost games*: each player minimizes a cost rather than maximizing a utility. This is equivalent to the usual utility convention by negating payoffs.

Definition 13.1 (Cost-form normal-form game)**.** Fix a finite player set $N = \{1, \dots, n\}$. A *(pure-strategy) cost game* is a tuple

$$G = \big((X_i)_{i\in N},\ (c_i)_{i\in N}\big)$$

where each X_i is a strategy space and each $c_i : \prod_{j\in N} X_j \to \overline{\mathbb{R}}$ is an extended-real cost function. Player i seeks to minimize $c_i(x_i, x_{-i})$ over $x_i \in X_i$ given the opponents' profile x_{-i}.

Definition 13.2 (Nash equilibrium and ε-Nash equilibrium)**.** Let G be a cost game with joint strategy space $X := \prod_i X_i$. A profile $x^\star \in X$ is a *(pure) Nash equilibrium* if for every player i,

$$c_i(x^\star) \ = \inf_{x_i' \in X_i} c_i(x_i', x_{-i}^\star).$$

More generally, for $\varepsilon \geq 0$, $x^\star$ is an *ε-Nash equilibrium* if for all i,

$$c_i(x^\star) \ \leq \inf_{x_i' \in X_i} c_i(x_i', x_{-i}^\star) + \varepsilon.$$

Existence of Nash equilibria typically requires compactness/convexity and the move to mixed strategies [47, 56]; we address mixed strategies below using Kleisli semantics.

13.2 A Nash-gap functional and reduction to single-objective optimization

A central bridge between games and optimization is that equilibrium conditions can be encoded by a *single* nonnegative functional whose global minima correspond to equilibria. For convex games this idea originates with the Nikaidô–Isoda gap function [52, 65] and is closely connected to variational inequality gap functions [16].

Definition 13.3 (Nikaidô–Isoda (Nash) gap)**.** Let G be a cost game and write $X = \prod_i X_i$. Define the *Nash gap* $\Gamma_G : X \to \overline{\mathbb{R}}$ by

$$\Gamma_G(x) \ := \sum_{i=1}^{n} \Big(c_i(x) \ - \inf_{x_i' \in X_i} c_i(x_i', x_{-i}) \Big).$$

Each summand is the *unilateral regret* of player i at x.

Lemma 13.4 (Gap characterizes (approximate) equilibria)**.** *Assume all infima in Definition 13.3 are finite.*

1. $\Gamma_G(x) \geq 0$ *for all* $x \in X$.
2. x *is a Nash equilibrium iff* $\Gamma_G(x) = 0$.
3. *If* x *is an* ε*-Nash equilibrium then* $\Gamma_G(x) \leq n\varepsilon$*. Conversely, if* $\Gamma_G(x) \leq \delta$ *then* x *is a* δ*-Nash equilibrium (in the sense of Definition 13.2) up to a factor* n *by distributing* δ *across players.*

Proof. Nonnegativity holds termwise because $c_i(x) \geq \inf_{x_i'} c_i(x_i', x_{-i})$. Equality $\Gamma_G(x) = 0$ holds iff each term vanishes, i.e. each player is a best response. The ε-bounds follow by summing playerwise regrets. □

Game-to-optimization translation. Definition 13.3 yields a canonical translation of equilibrium computation to optimization: define the (single-objective) optimization problem

$$\mathcal{N}(G) \;:=\; \big(X,\, \Gamma_G\big) \in \mathbf{OptRed}_{\mathrm{ex}},$$

so that $\mathsf{Sol}(\mathcal{N}(G))$ is precisely the set of Nash equilibria of G by Lemma 13.4. This identifies Nash equilibrium computation as a special case of the solution functor Sol developed in Sections 3–4.

13.3 A category of game reductions compatible with $\mathbf{OptRed}^{\mathrm{gr}}_{T,\alpha}$

We now define a category of games in which morphisms are *certified reductions* between equilibrium problems, mirroring Definition 8.3 and the unified grading of Section 10.

Definition 13.5 (Nash-gap reductions)**.** Let G and H be cost games with joint strategy spaces X_G and X_H and Nash gaps Γ_G and Γ_H. A *graded Nash-gap reduction* $a : G \to H$ consists of

$$\mathrm{enc}_a : X_G \to T(X_H), \qquad \mathrm{dec}_a : X_H \to T(X_G), \qquad g_a = (\alpha_a, \beta_a, w_a) \in (\mathbf{Aff}_{\geq 0} \times W),$$

where T is an effect monad (take $T = \mathrm{id}$ for deterministic reductions), together with the two inequalities expressed via the Kleisli lifting $(-)^\dagger$ (Section 7):

$$\mathrm{enc}_a^\dagger(\Gamma_H) \leq \Gamma_G, \tag{11}$$

$$\mathrm{dec}_a^\dagger(\Gamma_G) \leq \alpha_a\, \Gamma_H + \beta_a. \tag{12}$$

Proposition 13.6 (Composition and the unified grade)**.** *Nash-gap reductions compose by Kleisli composition on* $(\mathrm{enc}, \mathrm{dec})$ *and by the unified monoid law on grades:*

$$(\alpha, \beta, w) \otimes (\alpha', \beta', w') = (\alpha\alpha',\ \beta + \alpha\beta',\ w \otimes_W w').$$

In particular, the class of cost games and Nash-gap reductions forms a category, denoted $\mathsf{GameRed}_{T,G}$.

Proof. This is a direct instantiation of the proofs for approximate reductions (Proposition 8.9) and resource grading (Proposition 9.3) applied to the single cost Γ_G. □

Proposition 13.7 (Integration with $\mathbf{OptRed}^{\mathrm{gr}}_{T,\alpha}$ via a faithful functor)**.** *There is a faithful functor*

$$\mathcal{N} : \mathsf{GameRed}_{T,G} \longrightarrow \mathbf{OptRed}^{\mathrm{gr}}_{T,\alpha}$$

that sends a game G to the optimization object (X_G, Γ_G) and sends a Nash-gap reduction $(\mathrm{enc}_a, \mathrm{dec}_a, g_a) : G \to H$ to the corresponding graded/effectful reduction $(\mathrm{enc}_a, \mathrm{dec}_a, g_a) : \mathcal{N}(G) \to \mathcal{N}(H)$ in $\mathbf{OptRed}^{\mathrm{gr}}_{T,\alpha}$. Under this embedding, Nash equilibria are exactly $\mathsf{Sol}(\mathcal{N}(G))$.

Remark 13.8. One can strengthen $\mathsf{GameRed}_{T,G}$ by requiring enc_a and dec_a to factor playerwise ($\mathrm{enc}_a = \prod_i \mathrm{enc}_{a,i}$), which captures strategy reparameterizations and per-player preprocessing. The Nash-gap formulation above is intentionally agnostic: it allows global encodings such as introducing auxiliary variables, duplicating coordinates, or converting constraints to penalties, provided they are certified by the gap inequalities.

13.4 Special classes: potential games, zero-sum games, and convex (Rosen) games

The translation $G \mapsto (X_G, \Gamma_G)$ is always available, but in important subclasses games reduce to standard optimization problems more directly.

Potential games as exact reductions to optimization. A (cost) game G is a *potential game* if there exists a potential function $\Phi : X \to \overline{\mathbb{R}}$ such that for every player i and all $x_i, x_i' \in X_i$ (with x_{-i} fixed),

$$\Phi(x_i', x_{-i}) - \Phi(x_i, x_{-i}) \;=\; c_i(x_i', x_{-i}) - c_i(x_i, x_{-i}).$$

In such games, unilateral improvements in player costs are exactly tracked by Φ; see [46]. Every local minimizer of Φ is a Nash equilibrium, and under mild regularity assumptions global minimizers are Nash equilibria as well. Categorically, this yields an exact reduction from the game to the optimization problem (X, Φ), often allowing one to reuse convex-optimization pipelines when Φ is convex.

Two-player zero-sum games as saddle-point optimization. Let $N = \{1, 2\}$ and suppose there exists $f : X_1 \times X_2 \to \mathbb{R}$ with

$$c_1(x_1, x_2) = f(x_1, x_2), \qquad c_2(x_1, x_2) = -f(x_1, x_2).$$

Then Nash equilibria coincide with saddle points of f and can be characterized by minimax theorems [51, 69]. When f is convex–concave on convex compact sets, equilibrium computation is a convex-concave saddle-point problem, solvable by primal–dual and extragradient-type methods that fit naturally into $\mathbf{OptRed}^{\mathrm{gr}}_{T,\alpha}$ with resource grades (iterations, oracle calls) and approximation grades (duality gaps).

Convex games and variational inequalities. For differentiable games where each $c_i(\cdot, x_{-i})$ is convex on a convex set X_i, Rosen's theory of concave games [65] (in utility form) and its modern extensions show that Nash equilibria can be characterized as solutions to a variational inequality. Define the pseudogradient

$$F(x) := \big(\nabla_{x_1} c_1(x), \ldots, \nabla_{x_n} c_n(x)\big),$$

and let $X = \prod_i X_i$. A point $x^\star \in X$ is a Nash equilibrium if it solves the variational inequality $\mathrm{VI}(X, F)$:

$$\langle F(x^\star), x - x^\star \rangle \geq 0 \qquad \forall x \in X.$$

Such problems admit *gap functions* $g : X \to \mathbb{R}_{\geq 0}$ (e.g. Minty or dual gaps) with $g(x^\star) = 0$ iff $x^\star$ solves $\mathrm{VI}(X, F)$ [16]. Therefore convex games embed into $\mathbf{OptRed}_{\mathrm{ex}}$ not only via Γ_G but also via VI gap-minimization problems; the choice of gap is itself a reduction in $\mathbf{OptRed}^{\mathrm{gr}}_{T,\alpha}$.

13.5 Randomization: mixed strategies, correlated equilibrium, and stochastic games

The effect layer (Section 7) is particularly natural for game theory because randomization is part of the standard definition of equilibrium.

Mixed strategies as $\mathcal{D}$-effects. For finite (or suitably measurable) X_i, a mixed strategy for player i is a distribution $\sigma_i \in \mathcal{D}(X_i)$. The product distribution $\sigma = \bigotimes_i \sigma_i$ induces expected costs

$$\bar{c}_i(\sigma) \;:=\; \mathbb{E}_{x \sim \sigma}\big[c_i(x)\big],$$

which is exactly the cost-algebra evaluation $\alpha = \mathbb{E}$ discussed in Section 7 [47, 24, 30]. In this view, "randomize your action" is literally a Kleisli arrow, and equilibrium conditions become best-response statements in the $\mathcal{D}$-Kleisli world.

Correlated equilibrium as convex feasibility/optimization. A correlated equilibrium is a joint distribution $\mu \in \mathcal{D}(X)$ satisfying linear incentive constraints [3, 56]. For finite games these constraints define a polytope, so computing a correlated equilibrium with a chosen objective (e.g. maximize social welfare) is a linear program. Thus the correlated equilibrium correspondence embeds into the convex-optimization subcategory in Section "Application-oriented instantiations", and approximation/resources can be tracked by the unified grade.

Stochastic games and kernel semantics. In stochastic (dynamic) games [68], the state evolves according to controlled transition probabilities. Let S be a state space and let $K : S \times A_1 \times \cdots \times A_n \to \mathcal{D}(S)$ be a Markov kernel. The dynamics are already morphisms in $\mathbf{Kl}(()\mathcal{D})$, and equilibrium concepts are fixed points of dynamic-programming operators (Shapley operators). This aligns with the random-field section above: stochastic games can be seen as optimization problems on random fields where payoffs are functionals of entire trajectories.

13.6 Compositional game theory and open games

Beyond normal-form games, category theory has been used to give genuinely *compositional* syntax and semantics for games, in which complex games are built by wiring together simpler components. Open games provide one such formalism [23, 26].

Open games (sketch). An open game $\mathcal{G} : (X, S) \to (Y, R)$ can be presented (in one common formulation) by a tuple

$$(\Sigma_{\mathcal{G}},\ P_{\mathcal{G}},\ C_{\mathcal{G}},\ \mathsf{BR}_{\mathcal{G}}),$$

where $\Sigma_{\mathcal{G}}$ is a set of strategy profiles, $P_{\mathcal{G}} : \Sigma_{\mathcal{G}} \times X \to Y$ is a play function, $C_{\mathcal{G}} : \Sigma_{\mathcal{G}} \times X \times R \to S$ is a "coplay" function propagating utilities backwards, and $\mathsf{BR}_{\mathcal{G}}$ is a best-response relation depending on a continuation. Open games compose sequentially and in parallel, forming a symmetric monoidal (double) category in which equilibria are states satisfying a best-response condition [23, 26].

Integration with optimization reductions. Each "decision node" in an open game is an *argmin/argmax* operation against a continuation payoff. But argmin is precisely the solution functor Sol (or its effectful/approximate analogs) applied to a parametrized optimization problem. Consequently, open games can be interpreted as *programs* that call an optimization solver as a subroutine, and semantics-preserving rewrites of open games (morphisms of open games) correspond to higher morphisms in the (∞, n) framework (Section 17). For probabilistic games, kernel-coupling equivalences provide a canonical class of weak equivalences W; for nondeterministic game semantics, relational isomorphisms play the same role; and for syntactic game descriptions, certified compiler transformations supply proof-relevant weak equivalences.

Takeaway. The Nash-gap translation $\mathcal{N}$ shows that equilibrium computation can be treated uniformly as an optimization problem within $\mathbf{OptRed}^{\mathrm{gr}}_{T,\alpha}$, while open-game formalisms show that *game composition* itself is categorical. Together, they support an integrated view: (i) games are systems of interacting optimization problems, (ii) equilibrium computation is optimization (often convex/VI/saddle-point), and (iii) compositional game syntax and semantics can be organized by the same higher-categorical machinery used above for optimization pipelines.

14 Random fields and general stochastic fields

Many optimization problems have *field-valued* decision variables (spatial or spatiotemporal functions) and/or are driven by stochastic forcing. Examples range from spatial statistics and inverse problems to stochastic PDE-constrained optimization and Bayesian optimal design. The categorification developed above extends naturally once we recognize that a random field is simply a random element in a (typically very large) configuration space, and that many standard field operations (restriction, coarse-graining, conditioning, simulation) are compositional transformations that fit the reduce–solve–recover pattern.

14.1 Random fields as random variables in configuration spaces

Fix an index set (or "domain") S and a value space V. A *(discretized) field configuration* is a function $\phi : S \to V$, i.e. an element of the product V^S. In the finitary setting adopted earlier for $T = \mathcal{D}$, we take S finite (a grid, mesh, or set of measurement locations) and interpret a *random field* as a probability distribution $\mu \in \mathcal{D}(V^S)$, equivalently as a Kleisli arrow

$$* \longrightarrow \mathcal{D}(V^S)$$

from a singleton set $*$. Deterministic field transforms are ordinary functions $V^S \to V^{S'}$, while stochastic transforms are Markov kernels $V^S \to \mathcal{D}(V^{S'})$ (Kleisli arrows), composed by Kleisli composition [45, 70, 30].

For general stochastic fields with infinite index sets (e.g. $S \subset \mathbb{R}^d$), one usually works with measures on function spaces and their finite-dimensional marginals. Conceptually, the same pattern holds if one replaces the finitary distribution monad on $\mathbf{Set}$ by the Giry monad on measurable spaces [24] and uses projective consistency (Kolmogorov extension) to pass between finite discretizations and a limiting field law [31]. The present paper keeps the finitary $\mathcal{D}$ picture for simplicity, but the categorical constructions are designed to be compatible with such measure-theoretic upgrades.

14.2 Structured subcategories: Gaussian fields and Markov random fields

In applications, random fields often carry additional structure that can be used to define useful full subcategories. Two canonical examples are:

- **Gaussian random fields (GRFs):** laws on V^S determined by mean and covariance. For continuous-index fields, regularity and geometry of sample paths are central [1]. Categorical morphisms between GRFs include linear transformations, marginalization/restriction, and conditioning, which correspond to well-known algebra on covariance operators.
- **Gaussian Markov random fields (GMRFs) and general MRFs:** laws with conditional independence structure encoded by a graph (sparsity of precision matrices in the Gaussian case) [66]. Morphisms include graph coarsening/refinement, factorization changes, and message-passing style kernels; weak equivalences can be taken to be factor-graph isomorphisms or coupling equivalences on the induced kernels (Section 17).

These structured subcategories matter because they determine what counts as an admissible "representation change" and what invariants (sparsity, stationarity, locality) should be preserved by reductions.

14.3 Field-valued optimization problems and risk aggregation

Field optimization problems fit directly into the base definition $P = (X, c_P)$ by taking X to be a configuration space (or a discretized approximation thereof). For instance, with $X = V^S$ one may consider energies of the form

$$c(\phi) \;=\; \sum_{s\in S} \psi_s(\phi(s)) \;+\; \sum_{(s,t)\in E} \psi_{st}(\phi(s), \phi(t)),$$

which includes pairwise MRF energies and many variational formulations.

Stochasticity enters in two common ways. First, the objective may depend on random data ξ (observations, forcing, random coefficients), yielding a random cost $\phi \mapsto c(\phi; \xi)$. In the Kleisli framework this is naturally a map

$$C : X \longrightarrow \mathcal{D}(\overline{\mathbb{R}}),$$

and a choice of T-algebra $\alpha : \mathcal{D}(\overline{\mathbb{R}}) \to \overline{\mathbb{R}}$ (expectation, worst-case, etc.) produces a deterministic aggregated cost $\alpha \circ C : X \to \overline{\mathbb{R}}$ as in Section 7. Second, the decision variable itself may be a *policy* mapping observations to actions, which can be modeled by enlarging the state space so that a candidate is a measurable map and the algorithmic morphisms act on such maps (e.g. dynamic programming as a reduction on Bellman operators).

14.4 Discretization, coarsening, and multilevel reductions

A major source of algorithmic structure for field problems is the existence of multiple discretizations/resolutions. Let $\mathcal{I}$ be a poset (viewed as a thin category) of discretizations, ordered by refinement. For each $i \in \mathcal{I}$ let X_i be the corresponding finite-dimensional approximation of the configuration space (e.g. nodal values on a mesh), and let $P_i = (X_i, c_i)$ be the induced optimization problem.

A *coarsening map* $r_{ij} : X_j \to X_i$ for $i \leq j$ (restriction/averaging) and a *prolongation map* $p_{ij} : X_i \to X_j$ (interpolation) define an exact or approximate reduction $P_j \to P_i$ (encode by restriction, decode by prolongation). Discretization error naturally appears as an approximation grade $\gamma = (\alpha, \beta)$ in the sense of Section 8, while resource grades capture the number of degrees of freedom and the cost of applying operators. Chains of discretizations

$$i_0 \leq i_1 \leq \cdots \leq i_k$$

therefore assemble into composable morphisms in $\mathbf{OptRed}^{\mathrm{gr}}_{T,\alpha}$, making multilevel methods (multigrid, multiscale optimization, coarse-to-fine initialization) literally functorial.

14.5 Stochastic field dynamics and inference kernels

Stochastic field models often come equipped with dynamics: a stochastic process $(\Phi_t)_{t\geq 0}$ with state space $X = V^S$ (or a path/field space) induces a Markov kernel $K_t : X \to \mathcal{D}(X)$ for each time step t, and discrete-time simulation corresponds to Kleisli composition. This includes lattice Markov chains (e.g. Gibbs samplers) and discretizations of continuous-time diffusions (e.g. Langevin dynamics) used for sampling and optimization [63, 54]. In the present framework:

- an MCMC transition is an effectful endomorphism $X \to \mathcal{D}(X)$, optionally graded by cost per step;
- coupling arguments (Section 17) become canonical higher morphisms witnessing equivalence or contraction properties of kernels; and
- approximate inference (variational families, sparse approximations, truncations) is naturally expressed as an approximate reduction between field problems, with grades tracking both approximation error and computational cost.

Takeaway. Random fields and stochastic fields do not require a new foundational move: they fit by taking "decision sets" to be configuration/path spaces and by using the same Kleisli/graded machinery to model stochastic transforms, discretization maps, and approximate inference/optimization pipelines. The higher-categorical viewpoint is particularly well-suited here because equivalences between stochastic field constructions are rarely strict equalities; they are witnessed by couplings, factorization isomorphisms, and certified rewrites that naturally live as higher morphisms.

15 Higher-categorical extensions: from $\mathrm{OptRed}^{\mathrm{gr}}_{T,\alpha}$ to n-categories

The previous sections deliberately worked 1-categorically: optimization problems are objects and reductions/algorithms are morphisms. In practice, however, *algorithms have structure beyond being arrows*: there are

multiple implementations of the same reduction, compiler rewrites between implementations, proof objects witnessing approximation/resource bounds, and sometimes continuous deformations (homotopies) between algorithms. These are naturally modeled by *higher* morphisms. This section sketches a principled upgrade of the present categorification to n-categories (and, by completion, to $(\infty, 1)$-categories) [6, 38].

15.1 A minimal locally posetal 2-category of reductions

A first (and often sufficient) 2-categorical refinement is to keep the same objects and 1-morphisms as in $\mathbf{OptRed}^{\mathrm{gr}}_{T,\alpha}$, but add 2-morphisms expressing *refinement* between parallel reductions. Technically, one asks that the ambient effectful world carry a compatible order so that "no worse than" is stable under composition.

Definition 15.1 (Order-enriched Kleisli arrows)**.** Fix a monad T on $\mathbf{Set}$. Suppose each set Y is equipped with a preorder $(TY, \sqsubseteq_Y)$ such that:

(i) for every function $h : Y \to Y'$, the map $Th : TY \to TY'$ is monotone;

(ii) the unit $\eta_Y : Y \to TY$ is monotone (with Y discrete);

(iii) Kleisli composition is monotone in both arguments: if $f \sqsubseteq f' : X \to TY$ and $g \sqsubseteq g' : Y \to TZ$ (pointwise), then $g \odot f \sqsubseteq g' \odot f'$.

Remark 15.2. For nondeterminism, $T = \mathcal{P}$ with $\sqsubseteq$ given by $\subseteq$, and (i)–(iii) hold. For probability, one can choose various stochastic orders on TY (e.g. convex order, first-order stochastic dominance) provided they are stable under pushforward and Kleisli composition; the present development is agnostic to the choice.

Definition 15.3 (Refinement 2-cells)**.** Let P, Q be objects of $\mathbf{OptRed}^{\mathrm{gr}}_{T,\alpha}$, and let $a, b : P \to Q$ be parallel morphisms in $\mathbf{OptRed}^{\mathrm{gr}}_{T,\alpha}$ with

$$a = (\mathrm{enc}_a, \mathrm{dec}_a, g_a), \qquad b = (\mathrm{enc}_b, \mathrm{dec}_b, g_b),$$

where $g_a, g_b \in G = \mathbf{Aff}_{\geq 0} \times W$ are grades. Define a *2-cell* $a \Rightarrow b$ to be the conjunction of the componentwise refinements

$$\mathrm{enc}_a \sqsubseteq \mathrm{enc}_b, \qquad \mathrm{dec}_a \sqsubseteq \mathrm{dec}_b, \qquad g_a \leq g_b,$$

where $\leq$ on G is the product preorder (componentwise) and $\sqsubseteq$ is the pointwise preorder on Kleisli arrows induced by Definition 15.1.

Proposition 15.4 (A strict 2-category of graded/effectful reductions)**.** *Under the hypotheses of Definition 15.1, the data above define a* strict *2-category* $\mathbf{OptRed}^{\mathrm{gr}}_{T,\alpha}{}^{\leq}$ *whose:*

- *objects are those of* $\mathbf{OptRed}^{\mathrm{gr}}_{T,\alpha}$*;*
- *1-morphisms are those of* $\mathbf{OptRed}^{\mathrm{gr}}_{T,\alpha}$*;*
- *2-morphisms are refinement 2-cells as above.*

Vertical and horizontal composition of 2-cells are induced by transitivity and monotonicity, and the interchange law holds automatically because the homs are preorders (so $\mathbf{OptRed}^{\mathrm{gr}}_{T,\alpha}{}^{\leq}$ *is locally posetal).*

Remark 15.5. This "minimal" 2-categorification is useful when 2-morphisms are meant to express only the *fact* that one pipeline refines another (e.g. tighter approximation and/or lower resources), not a concrete witness. However, because the hom-categories are thin, there is no room for nontrivial 3-morphisms: to get genuinely higher structure one must make 2-cells proof-relevant.

15.2 Proof-relevant 2-cells and bicategorical semantics

To keep track of *witnesses* for refinement/bounds, it is natural to work in a bicategory (weak 2-category) [6]. Concretely, instead of taking 1-morphisms to live in $\mathbf{Kl}(()T)$ (a 1-category), one chooses an ambient bicategory $\mathcal{B}$ whose 1-cells model effectful computations and whose 2-cells model *transformations* between computations (e.g. maps of spans for nondeterminism, couplings for stochastic kernels, or program rewrites).

Definition 15.6 (Optimization reductions internal to a bicategory)**.** Let $\mathcal{B}$ be a bicategory. Define a bicategory $\mathbf{OptRed}(\mathcal{B})$ as follows.

- Objects are optimization problems $P = (X, c_P)$ where X is an object of $\mathcal{B}$ and c_P is an external cost map on the "states" of X (as in earlier sections).

- A 1-cell $a : P \to Q$ is reduction data $(\mathrm{enc}_a, \mathrm{dec}_a, g_a)$ where $\mathrm{enc}_a : X \to Y$ and $\mathrm{dec}_a : Y \to X$ are 1-cells of $\mathcal{B}$ and $g_a \in G$ is a grade, subject to the same cost/approximation constraints as in $\mathbf{OptRed}^{\mathrm{gr}}_{T,\alpha}$ (interpreted in $\mathcal{B}$ via an appropriate evaluation functor).
- A 2-cell $\theta : a \Rightarrow b$ between parallel 1-cells $a, b : P \to Q$ is a pair of 2-cells in $\mathcal{B}$,

$$\theta_{\mathrm{enc}} : \mathrm{enc}_a \Rightarrow \mathrm{enc}_b, \qquad \theta_{\mathrm{dec}} : \mathrm{dec}_a \Rightarrow \mathrm{dec}_b,$$

  together with a grade comparison $g_a \leq g_b$ in G.

Remark 15.7. The ambient bicategory $\mathcal{B}$ can be chosen to match the computational interpretation:

- *Nondeterminism:* relations $X \to Y$ may be modeled as spans $X \leftarrow A \to Y$ in $\mathbf{Set}$; then 2-cells are maps of spans, giving proof-relevant witnesses rather than bare inclusions.
- *Randomness:* stochastic maps can be modeled in bicategories of Markov kernels; 2-cells can be chosen as appropriate couplings or as explicit rewrites between kernel descriptions.

In either case, vertical and horizontal compositions are inherited from $\mathcal{B}$, and one obtains a genuine bicategory of optimization reductions.

15.3 From bicategories to n-categories

Once the homs are categories (or bicategories), the step to higher categories is systematic: one iterates the idea "morphisms have morphisms between them" [38]. There are two standard routes.

(A) Enrichment in $(n-1)$-categories. For each pair of problems P, Q, let $\mathsf{Red}(P, Q)$ be an $(n-1)$-category whose 0-cells are reductions $P \to Q$, 1-cells are rewrites/witnesses between reductions, 2-cells are rewrites between rewrites, and so on up to level $n-2$. If composition of reductions extends to an $(n-1)$-functor

$$\mathsf{Red}(Q, R) \times \mathsf{Red}(P, Q) \longrightarrow \mathsf{Red}(P, R)$$

satisfying the usual (weak) associativity/unit coherences, the resulting structure is a weak n-category $\mathbf{OptRed}^{(n)}_{\mathrm{ex}}$ of optimization problems.

(B) $(\infty, 1)$-categorical completion and truncation. Even if one only constructs a 2-category (or bicategory) $\mathcal{O}$ of problems and reductions, one can form an $(\infty, 1)$-category by taking nerves of the hom-categories and assembling them into mapping spaces. For instance, if each hom $\mathcal{O}(P, Q)$ is an ordinary category, its nerve $N(\mathcal{O}(P, Q))$ is a simplicial set; under mild functoriality hypotheses this yields a simplicially enriched category whose homotopy coherent nerve is an $(\infty, 1)$-category [42, 61]. The n-categorical truncation then keeps k-morphisms only up to level n (or, equivalently, truncates mapping spaces).

15.4 Higher natural transformations

In a 2-category (or higher), the functorial constructions of Sections 3–5 naturally upgrade: functors become 2-functors (or n-functors), natural transformations become 2-morphisms, and one additionally has *modifications* (3-morphisms) between natural transformations, and higher analogues thereafter [6, 38]. For example:

- The inclusions $\iota : \mathsf{Feas} \Rightarrow \mathsf{Cand}$ and $\kappa : \mathsf{Sol} \Rightarrow \mathsf{Feas}$ may be viewed as 2-cells between 2-functors when $\mathsf{Feas}, \mathsf{Cand}, \mathsf{Sol}$ take values in an ambient 2-category (e.g. a bicategory of computations or proof objects).
- The "cost as oplax natural transformation" viewpoint (Section 5) is already 2-categorical: $\mathbf{Pos}$ is a locally posetal 2-category, and oplaxity is precisely a statement about 2-cells.

15.5 Higher Kleisli objects and graded effects

Effects were encoded 1-categorically via a monad T and its Kleisli category. In a bicategory (or 2-category), monads generalize to *pseudomonads*, and the correct analogue of the Kleisli construction is a *Kleisli object* in the sense of the formal theory of monads [70]. Thus, to extend the present framework to higher categories, one may:

(i) choose an ambient n-category (or $(\infty, 1)$-category) $\mathcal{B}$ modeling computations;

(ii) equip it with an appropriate (pseudo)monad encoding randomness/nondeterminism;

(iii) form the corresponding Kleisli n-category (when it exists);

(iv) internalize grades by a writer-style construction in $\mathcal{B}$ (a graded/parameterised effect), generalizing Section 9 [2, 32, 55].

The 1-categorical categories $\mathbf{OptRed}_{T,\alpha}, \mathbf{OptRed}_{\approx}, \mathbf{OptRed}_{\mathrm{res}}, \mathbf{OptRed}^{\mathrm{gr}}_{T,\alpha}$ are recovered by truncating the higher structure to 0- and 1-morphisms and forgetting proof objects.

16 Infinity-categorical extensions

The higher-categorical discussion of Section 9 produced (weak) n-categorical refinements by making algorithmic correctness and rewrite data *proof-relevant*. A further natural step is to pass from finite-dimensional higher categories to ∞-categories, in which all coherence data is retained systematically (rather than truncated at a fixed height). The basic idea is to start with a *relative* category of reductions—a category equipped with a chosen class of "weak equivalences"—and then apply a standard localization construction to obtain an $(\infty, 1)$-category of optimization problems and pipelines [42, 14].

16.1 Relative categories of reductions and simplicial localization

Let $\mathbf{OptRed}^{\mathrm{gr}\ \mathrm{pr}}_{T,\alpha}$ be a *proof-relevant* version of the unified category $\mathbf{OptRed}^{\mathrm{gr}}_{T,\alpha}$ (Section 8). Concretely, $\mathbf{OptRed}^{\mathrm{gr}\ \mathrm{pr}}_{T,\alpha}$ has the same objects (optimization problems) and the same 1-morphisms (graded/effectful/approximate reductions), but now we assume that for each pair of problems P, Q the collection of reductions $P \to Q$ carries a category (or groupoid) of *algorithmic equivalences and rewrites*. For example, one may take:

- objects of $\mathbf{OptRed}^{\mathrm{gr}\ \mathrm{pr}}_{T,\alpha}(P, Q)$ to be reductions $a : P \to Q$;
- morphisms $a \to b$ to be *invertible* rewrites witnessing that a and b implement the same reduction (e.g. a semantics-preserving compiler transformation, a coupling of stochastic kernels, or a proof that two derived bounds are equal).

The precise choice depends on what one wishes to identify "up to"; the only requirement for the $(\infty, 1)$-construction below is that higher comparison data be invertible (groupoid-like).

Let $W \subseteq \mathrm{Mor}(\mathbf{OptRed}^{\mathrm{gr}\ \mathrm{pr}}_{T,\alpha})$ be a distinguished class of *weak equivalences* (for instance, those reductions whose grade is the unit $(1, 0, I) \in \mathbf{Aff}_{\geq 0} \times W$ and which admit a two-sided inverse up to invertible rewrite). The pair $(\mathbf{OptRed}^{\mathrm{gr}\ \mathrm{pr}}_{T,\alpha}, W)$ is then a *relative category*. Its simplicial localization $L_W(\mathbf{OptRed}^{\mathrm{gr}\ \mathrm{pr}}_{T,\alpha})$ is a simplicial category (i.e. a category enriched in simplicial sets) whose homotopy category is the ordinary localization $\mathbf{OptRed}^{\mathrm{gr}\ \mathrm{pr}}_{T,\alpha}[W^{-1}]$ [14]. Passing to the homotopy coherent nerve yields an $(\infty, 1)$-category

$$\mathcal{O}_\infty \ := \ N_{\mathrm{hc}}\big(L_W(\mathbf{OptRed}^{\mathrm{gr}\ \mathrm{pr}}_{T,\alpha})\big),$$

well-defined up to equivalence of ∞-categories [42, 61].

Mapping spaces as "spaces of pipelines". For problems P, Q, the mapping space $\mathrm{Map}_{\mathcal{O}_\infty}(P, Q)$ is a simplicial set whose 0-simplices are reductions $P \to Q$, whose 1-simplices are (invertible) rewrites between reductions, and whose higher simplices encode coherent higher homotopies between such rewrites. Composition of pipelines is associative only up to coherent higher homotopy in $\mathcal{O}_\infty$, and the simplicial structure records these coherences automatically. An equivalent description is available in the complete Segal space model, using the Rezk nerve of a relative category [59].

16.2 Functorial constructions and naturality in ∞-categories

Many of the deterministic/Kleisli constructions in Sections 3–8 can be upgraded by the universal property of ∞-categorical localization. If $\mathcal{D}$ is an $(\infty, 1)$-category and $F : \mathbf{OptRed}^{\mathrm{gr}\ \mathrm{pr}}_{T,\alpha} \to \mathcal{D}$ is a functor that sends every weak equivalence in W to an equivalence in $\mathcal{D}$, then F factors (essentially uniquely) through the localization:

$$\mathbf{OptRed}^{\mathrm{gr}\ \mathrm{pr}}_{T,\alpha} \longrightarrow \mathcal{O}_\infty \xrightarrow{\widetilde{F}} \mathcal{D},$$

and the space of such factorizations is contractible [42, §1.3.4]. In particular:

- The "candidate" and "solution" functors (e.g. $\mathsf{Cand}_T, \mathsf{Feas}_T, \mathsf{Sol}_T$) may be regarded as ∞-functors into an appropriate target ∞-category of computations.
- A natural transformation between ∞-functors is a 1-morphism in the functor ∞-category $\mathrm{Fun}(\mathcal{O}_\infty, \mathcal{D})$, while higher homotopies between natural transformations are higher simplices in the same mapping space [42, 61].

Thus "naturality" statements from earlier sections become literal equalities of 1-morphisms (or homotopies between them) in $\mathrm{Fun}(\mathcal{O}_\infty, \mathcal{D})$.

16.3 Effect monads, Kleisli ∞-categories, and graded effects

In the $(\infty, 1)$-world, monads are most naturally treated as associative algebra objects in the monoidal ∞-category of endofunctors (with tensor given by composition) [70, 43]. Under mild presentability/accessibility hypotheses on the ambient ∞-category of states, such monads admit Kleisli objects, producing a Kleisli ∞-category that generalizes the ordinary Kleisli category used in Section 5 [43, 61]. The effectful reduction data $\mathrm{enc}, \mathrm{dec}$ can then be interpreted as 1-morphisms in this Kleisli ∞-category.

The unified grade monoid $G = \mathbf{Aff}_{\geq 0} \times W$ (Section 8) can be viewed as a (discrete) monoidal ∞-category BG with one object and endomorphisms G. A grade-tracking mechanism is then simply an ∞-functor $\mathsf{Grade} : \mathcal{O}_\infty^{op} \to BG$. Equivalently, one may internalize grades as a writer-style effect: the endofunctor $(-) \times G$ is a monad, and the writer transformer combines it with the computational effect monad T to form a graded effect monad T^G, as in Section 8, now at the level of ∞-categories [72, 40, 55, 43].

16.4 Remark: beyond $(\infty, 1)$

An $(\infty, 1)$-category is appropriate when all higher comparison data is invertible ("homotopy-like"). If one wishes to retain *non-invertible* 2-cells such as refinement relations (e.g. "algorithm a dominates b"), then the correct ambient setting is an $(\infty, 2)$-category (or more generally an (∞, n)-category) rather than an $(\infty, 1)$-category. Section 17 gives a concrete (∞, n) enhancement with canonical domain-specific choices of weak equivalences. The locally posetal 2-categories of Sections 8–9 may be regarded as 2-truncated shadows of such higher structures.

17 (∞, n)-categorical extensions and canonical weak equivalences

Section 15 explained how to enrich the 1-category of reductions by adding proof-relevant higher morphisms, and Section 16 constructed an $(\infty, 1)$-localization by inverting a chosen class of weak equivalences. Many optimization settings, however, involve *non-invertible* higher comparison data (refinement, simulation, certified but one-way compiler passes, etc.), while still demanding that all coherences above some finite height be invertible "up to homotopy". This is precisely the remit of (∞, n)-categories: k-morphisms for $k > n$ are invertible, but for $1 \leq k \leq n$ they need not be [5, 60].

17.1 From relative n-categories of reductions to an (∞, n)-category

There are many equivalent models of (∞, n)-categories (Segal n-categories, complete Θ_n-spaces, n-fold complete Segal spaces, etc.), and the correct attitude is to work model-independently whenever possible. The unicity theorem of Barwick–Schommer-Pries shows that the standard models present the same homotopy theory (up to a controlled $(\mathbb{Z}/2)^n$-ambiguity) [5]. For concreteness we speak in the language of complete Θ_n-spaces [60], but any equivalent model may be substituted.

Definition 17.1 (Proof-relevant n-category of reductions)**.** Fix $n \geq 2$. A *proof-relevant n-category of graded reductions* $\mathbf{OptRed}^{\mathrm{gr}}_{T,\alpha}{}^{(n)}$ has:

- objects: optimization problems $P = (X, c_P)$;
- 1-morphisms: graded/effectful/approximate reductions $a : P \to Q$ as in Definition 10.3;
- k-morphisms ($2 \leq k \leq n$): k-fold proof objects comparing reductions, e.g. refinement witnesses, compiler rewrites, couplings, and higher coherences between such data.

The only requirement is that the n-categorical compositions recover the 1-categorical composition on $\mathbf{OptRed}^{\mathrm{gr}}_{T,\alpha}$ at the level of 1-morphisms, and that all coherence data is supplied up to height n (cf. bicategories in Section 15).

Definition 17.2 (Weak equivalences)**.** A *choice of weak equivalences* on $\mathbf{OptRed}^{\mathrm{gr}}_{T,\alpha}{}^{(n)}$ is a wide sub-n-category $W \subseteq \mathbf{OptRed}^{\mathrm{gr}}_{T,\alpha}{}^{(n)}$ whose 1-morphisms are to be formally inverted. Typically one requires:

(i) W contains all identity 1-morphisms;

(ii) W is closed under composition of 1-morphisms;

(iii) (2-out-of-3) whenever $f, g, g \circ f$ are composable 1-morphisms and two of them lie in W, so does the third, up to equivalence controlled by higher cells.

In our optimization setting it is natural (but not mandatory) to restrict to *unit-grade* equivalences: a 1-morphism a may lie in W only if its grade is the monoidal unit $e_G = (1, 0, I) \in \mathbf{Aff}_{\geq 0} \times W$.

Proposition 17.3 (Localization to an (∞, n)-category)**.** *Given* $(\mathbf{OptRed}^{\mathrm{gr}}_{T,\alpha}{}^{(n)}, W)$ *as above, there is an associated* (∞, n)*-category*

$$\mathcal{O}_{(\infty,n)} \;:=\; \mathsf{Loc}_W(\mathbf{OptRed}^{\mathrm{gr}}_{T,\alpha}{}^{(n)}),$$

obtained by applying an (∞, n)*-localization / Rezk completion (e.g. via the* Θ_n*-space Rezk nerve). Its homotopy* $(n, 1)$*-category has the same objects and 1-morphisms as* $\mathbf{OptRed}^{\mathrm{gr}}_{T,\alpha}{}^{(n)}$*, but with 1-morphisms in* W *inverted up to higher equivalence [60, 5].*

Remark 17.4 (Mapping $(\infty, n$–$1)$-categories)**.** For problems P, Q, the mapping object $\mathrm{Map}_{\mathcal{O}_{(\infty,n)}}(P, Q)$ is an $(\infty, n$–$1)$-category whose 0-cells are pipelines $P \to Q$, whose 1-cells are rewrites/refinements between pipelines, and so on up to level $n-1$. This is the precise higher-categorical analogue of "spaces of pipelines" from Section 16.1.

17.2 Canonical choices of W in three domains

The abstract definition of W becomes useful only after fixing what counts as "the same computation" or "the same representation change" in the domain of interest. We record three canonical choices matching the effect monads used throughout this note.

17.2.1 (i) Randomness: $\mathcal{D}$ and kernel–coupling equivalences

Work in the finitary setting where a randomized computation is a Kleisli arrow $k : X \to \mathcal{D}(Y)$ (a discrete Markov kernel). A *coupling* of distributions $\mu, \nu \in \mathcal{D}(Y)$ is a distribution $\pi \in \mathcal{D}(Y \times Y)$ whose marginals are μ and ν; see [41]. Pointwise couplings of kernels are defined similarly.

Definition 17.5 (Kernel coupling and diagonal coupling)**.** Let $k, k' : X \to \mathcal{D}(Y)$. A *(pointwise) coupling* of k and k' is a function

$$\pi : X \to \mathcal{D}(Y \times Y)$$

such that for each $x \in X$ the first marginal of $\pi(x)$ is $k(x)$ and the second marginal is $k'(x)$. A coupling π is *diagonal* if $\pi(x)$ is supported on the diagonal $\Delta_Y = \{(y, y) : y \in Y\}$ for all x. Diagonal couplings exist iff $k(x) = k'(x)$ for all x.

Diagonal couplings are therefore a canonical *proof-relevant equality* notion for kernels: instead of identifying two kernels by definitional equality, one records an explicit diagonal coupling witness. This perspective aligns with categorical probability frameworks in which Markov kernels are treated as morphisms [24, 30, 21].

Definition 17.6 (Coupling-equivalence and the class $W_{\mathcal{D}}$)**.** Let $\mathcal{K}_{\mathcal{D}}$ be the (weak) 2-category whose objects are sets, whose 1-morphisms are Markov kernels $X \to \mathcal{D}(Y)$, and whose 2-morphisms are diagonal couplings as in Definition 17.5. Call a kernel $k : X \to \mathcal{D}(Y)$ a *coupling-equivalence* if there exists a kernel $\ell : Y \to \mathcal{D}(X)$ and diagonal couplings

$$\ell \odot k \;\Rightarrow\; \eta_X, \qquad k \odot \ell \;\Rightarrow\; \eta_Y,$$

satisfying the usual triangle identities up to higher (invertible) coherence. Define $W_{\mathcal{D}}$ to be the class of graded reductions $a : P \to Q$ in $\mathbf{OptRed}^{\mathrm{gr}}_{T,\alpha}{}^{(n)}$ whose grade is e_G and whose encoding/decoding components are coupling-equivalences in $\mathcal{K}_{\mathcal{D}}$.

Remark 17.7. In the finitary $\mathcal{D}$ Kleisli category, the only strictly invertible kernels are deterministic bijections (permutation matrices). Definition 17.6 should therefore be read as: *invertible representation changes are those that are almost-surely bijective*, but we record their equality/zigzag laws via coupling witnesses.

17.2.2 (ii) Nondeterminism: $\mathcal{P}$ and relational isomorphisms

For $T = \mathcal{P}$, Kleisli arrows $X \to \mathcal{P}(Y)$ correspond to relations $R \subseteq X \times Y$ and Kleisli composition is relational composition [45, 19].

Definition 17.8 (Relational isomorphism and the class $W_{\mathcal{P}}$)**.** A relation $R \subseteq X \times Y$ is a *relational isomorphism* if there exists a relation $S \subseteq Y \times X$ such that (under relational composition) $S \circ R = \mathrm{id}_X$ and $R \circ S = \mathrm{id}_Y$. Equivalently, R is the graph of a bijection $X \cong Y$ [19]. Define $W_{\mathcal{P}}$ to be the class of graded reductions $a : P \to Q$ in $\mathbf{OptRed}^{\mathrm{gr}}_{T,\alpha}{}^{(n)}$ of unit grade whose encoding/decoding relations are relational isomorphisms.

17.2.3 (iii) Syntactic algorithms: certified compiler transformations

Finally, suppose reductions are implemented in a *syntactic* algorithm language $\mathcal{L}$ whose programs have a formal denotational or operational semantics $[\![-]\!]$ into the effectful world of interest (e.g. into $\mathbf{Kl}(T)$). A *certified compiler transformation* is a rewrite step $p \Rightarrow q$ equipped with a machine-checkable certificate that $[\![p]\!] = [\![q]\!]$ (semantic preservation). Such certificates are the core idea of formally verified compilers such as CompCert [39].

Definition 17.9 (Certified transformations as weak equivalences)**.** Fix input/output state sets (or types) X, Y. Let $\mathrm{Prog}(X, Y)$ be the category whose objects are $\mathcal{L}$-programs $p : X \to Y$ and whose morphisms are certified compiler transformations $p \Rightarrow q$. Let $W_{\mathrm{cc}}(X, Y)$ be the class of all certified transformations in $\mathrm{Prog}(X, Y)$. Then the localization $\mathrm{Prog}(X, Y)[W_{\mathrm{cc}}(X, Y)^{-1}]$ identifies programs that are connected by certified transformations, and can be upgraded to an $(\infty, 1)$-groupoid of programs by taking the simplicial localization. In the optimization setting, taking the hom-$(n-1)$-categories $\mathsf{Red}(P, Q)$ of Definition 17.1 to be built from such $\mathrm{Prog}(X, Y)$, we obtain a canonical choice of weak equivalences W generated by certified compiler transformations on the enc and dec components (and optionally respecting grades).

Remark 17.10 (What this buys you)**.** Localizing at certified compiler transformations ensures that the resulting (∞, n)-category remembers pipelines *up to certified compilation*: optimization reductions can be reasoned about at the level of semantics, while still retaining (non-invertible) refinement 2-cells and resource/approximation grades below height n.

18 Conclusion and outlook

We have treated the ubiquitous reduce–solve–recover pattern in optimization as a compositional mathematical object. Starting from a deterministic category of *exact* reductions between extended-real optimization problems [64], we made standard structural facts functorial: feasible sets, candidate solutions, and solution recovery arise as functors (often contravariantly), and the canonical inclusions among them are expressed as natural transformations [44, 4]. We then generalized the semantics along three axes that are unavoidable in practice:

- **Effects:** nondeterminism and randomness are handled by moving from $\mathbf{Set}$ to a Kleisli category of a monad [45, 70], with standard instances such as the powerset monad and distribution monads [24, 30]. Many of the deterministic constructions reappear as Kleisli-natural structure.
- **Approximation:** correctness is weakened to multiplicative/additive slack and packaged into an affine grade monoid, yielding systematic composition rules for approximation loss.
- **Resources:** runtime/memory bounds are tracked compositionally via monoidal/enriched structure [33, 37].

These threads were unified by a single graded/effectful framework in which each pipeline carries one grade encoding both quality loss and resource use, either externally as annotations or internally via a writer-style transformer. Finally, we lifted the entire picture from ordinary categories to higher categorical settings, arguing that higher morphisms naturally model algorithmic rewrites and correctness witnesses, and that (∞, n)-localization organizes the resulting coherences into mapping objects with genuine homotopy-theoretic content [38, 42, 5].

Concrete semantics via weak equivalences. A higher-categorical framework is only as meaningful as its notion of weak equivalence. We therefore proposed canonical domain-specific choices: kernel–coupling equivalences for probabilistic kernels [41, 21], relational isomorphisms for nondeterminism [19], and certified compiler transformations for syntactic algorithm languages [39]. These choices let one interpret mapping spaces as "spaces of pipelines" where paths are proof-relevant rewrites and higher simplices encode coherent commutations.

The payoff of the categorified viewpoint is not a new solver, but a structured language for building and comparing solver pipelines—one where correctness, approximation loss, and resources compose by design, and where equivalences between pipelines carry coherent higher structure. This is a forward path toward libraries of reusable, certified

reductions and transformations, and toward optimization toolchains whose "plumbing" is as principled as their core numerical methods.

19 Preliminary Approaches to Open Problems

This section outlines preliminary approaches to solving some open problems, specifically addressing challenges related to non-invertible higher structure, duality and relaxation theory, and mechanization and verification. These approaches offer starting points for further exploration and formalization within the categorical framework for optimization.

19.1 Non-invertible Higher Structure

Non-invertible higher structure arises because many optimization algorithms involve non-invertible refinements (e.g., local search methods, simulated annealing, and heuristics) rather than exact transformations. In classical category theory, invertible morphisms play a central role, but in optimization, many transformations are non-invertible yet still composable.

Approach: We propose to model non-invertible higher morphisms using weak morphisms or profunctors in an (∞, n)-category framework. These allow us to define transformations between optimization algorithms that do not require strict invertibility but still respect compositionality. For instance, a non-invertible 2-morphism could represent a refinement or dominance relation between two optimization methods, where one method is guaranteed to perform better than another but is not equivalent in a strict sense.

Enriched Categories: By using enriched categories, we can track degrees of refinement, representing how one algorithm dominates another, without requiring inverses. The natural composition of these weak morphisms will preserve the original structure while accommodating non-invertible transformations.

Tools: We will utilize profunctorial semantics and homotopy theory to formalize and compose non-invertible higher morphisms, providing a coherent way to capture such relations.

19.2 Duality and Relaxation Theory

Duality and relaxation play critical roles in optimization, particularly in convex optimization. While duality is well-understood for convex problems, it remains an open challenge to formalize duality and relaxation systematically within a categorical framework, especially for non-convex and approximate optimization.

Approach: We aim to model duality using adjoint functors. In convex optimization, primal and dual problems can be formalized as adjoint functors between categories, where one functor represents the primal problem and the other the dual. We propose to treat these adjunctions as contravariant endofunctors on the subcategory of optimization problems where regularity conditions (e.g., Slater's conditions) hold.

Relaxation can be treated as an approximate morphism where the relaxed problem is mapped to the original problem via a graded morphism. The grade will capture the degree of relaxation, and the morphism will track the approximation error introduced by the relaxation procedure.

We also aim to define optimality certificates that quantify the error between the relaxed and original solutions. These certificates will be formalized as graded morphisms that encode the gap between the solution and the relaxed version.

Tools: Adjoint functors, limit constructions, and profunctorial semantics will be used to formalize duality and relaxation theory, allowing us to capture these relationships in a rigorous, compositional way.

19.3 Mechanization and Verification

As optimization algorithms become more complex and involve approximations, ensuring their correctness becomes increasingly important. Mechanizing the optimization process and verifying its correctness within the categorical framework is a significant challenge, particularly when dealing with heuristic and approximate methods.

Approach: We propose linking the categorical framework with proof assistants and formal verification tools (e.g., Coq, Lean, Isabelle) to provide certified reductions. Each morphism in the category will be associated with a proof object that certifies the correctness of the algorithm. This will ensure that the transformations between problems are correct and that any approximations or relaxations are rigorously validated.

We can use dependent types to encode the approximation error, resource bounds, and correctness conditions in a formal system, allowing us to automate the verification of optimization algorithms and their transformations. Furthermore, certified compilers will play a key role in ensuring that optimizations on algorithms are semantics-preserving.

Tools: Coq, Lean, and Isabelle will be used to automate the verification of optimization algorithms within the categorical framework, ensuring that correctness is maintained at every step of the transformation process.

These preliminary approaches provide a starting point for addressing the open problems in categorifying optimization, duality, stochastic processes, and verification. Future work will refine these ideas, using categorical structures to develop a fully mechanized, verified, and unified framework for optimization algorithms, stochastic processes, and their transformations.

References

[1] Robert J. Adler and Jonathan E. Taylor. *Random Fields and Geometry*. Springer Monographs in Mathematics. Springer, 2007. DOI: 10.1007/978-0-387-48116-6.

[2] Robert Atkey. "Parameterised Notions of Computation". In: *Journal of Functional Programming* 19.3–4 (2009), pp. 335–376. DOI: 10.1017/S095679680900728X.

[3] Robert J. Aumann. "Subjectivity and Correlation in Randomized Strategies". In: *Journal of Mathematical Economics* 1.1 (1974), pp. 67–96. DOI: 10.1016/0304-4068(74)90037-8.

[4] Steve Awodey. *Category Theory*. 2nd ed. Oxford University Press, 2010. ISBN: 9780199587360.

[5] Clark Barwick and Christopher Schommer-Pries. "On the Unicity of the Theory of Higher Categories". In: *Journal of the American Mathematical Society* 34.4 (2021), pp. 1011–1058. DOI: 10.1090/jams/972.

[6] Jean Bénabou. "Introduction to Bicategories". In: *Reports of the Midwest Category Seminar*. Vol. 47. Lecture Notes in Mathematics. Springer, 1967, pp. 1–77.

[7] Dimitri P. Bertsekas. *Nonlinear Programming*. 2nd ed. Athena Scientific, 1999. ISBN: 9781886529007.

[8] Léon Bottou, Frank E. Curtis, and Jorge Nocedal. "Optimization Methods for Large-Scale Machine Learning". In: *SIAM Review* 60.2 (2018), pp. 223–311. DOI: 10.1137/16M1080173.

[9] Stephen Boyd and Lieven Vandenberghe. *Convex Optimization*. Cambridge University Press, 2004. ISBN: 9780521833783.

[10] Leo Breiman. "Bagging Predictors". In: *Machine Learning* 24.2 (1996), pp. 123–140. DOI: 10.1007/BF00058655.

[11] Leo Breiman. "Random Forests". In: *Machine Learning* 45.1 (2001), pp. 5–32. DOI: 10.1023/A:1010933404324.

[12] George B. Dantzig. *Linear Programming and Extensions*. Princeton University Press, 1963.

[13] Thomas G. Dietterich. "Ensemble Methods in Machine Learning". In: *Multiple Classifier Systems*. Vol. 1857. Lecture Notes in Computer Science. Springer, 2000, pp. 1–15. DOI: 10.1007/3-540-45014-9_1.

[14] William G. Dwyer and Daniel M. Kan. "Simplicial Localizations of Categories". In: *Journal of Pure and Applied Algebra* 17.3 (1980), pp. 267–284. DOI: 10.1016/0022-4049(80)90049-3.

[15] Agoston E. Eiben and James E. Smith. *Introduction to Evolutionary Computing*. Springer, 2003. DOI: 10.1007/978-3-662-05094-1.

[16] Francisco Facchinei and Jong-Shi Pang. *Finite-Dimensional Variational Inequalities and Complementarity Problems*. Springer Series in Operations Research and Financial Engineering. Springer, 2003. DOI: 10.1007/b97543.

[17] Brendan Fong and David I. Spivak. *An Invitation to Applied Category Theory: Seven Sketches in Compositionality*. Cambridge University Press, 2019. ISBN: 9781108711821.

[18] Yoav Freund and Robert E. Schapire. "A Decision-Theoretic Generalization of On-Line Learning and an Application to Boosting". In: *Journal of Computer and System Sciences* 55.1 (1997), pp. 119–139. DOI: 10.1006/jcss.1997.1504.

[19] Peter J. Freyd and Andre Scedrov. *Categories, Allegories*. Vol. 39. North-Holland Mathematical Library. North-Holland, 1990. ISBN: 0444703683.

[20] Jerome H. Friedman. "Greedy Function Approximation: A Gradient Boosting Machine". In: *The Annals of Statistics* 29.5 (2001), pp. 1189–1232. DOI: 10.1214/aos/1013203451.

[21] Tobias Fritz. *A Synthetic Approach to Markov Kernels, Conditional Independence and Theorems on Sufficient Statistics*. arXiv preprint. 2019. arXiv: 1908.07021 [math.ST].

[22] Michael R. Garey and David S. Johnson. *Computers and Intractability: A Guide to the Theory of NP-Completeness*. W. H. Freeman, 1979. ISBN: 9780716710455.

[23] Neil Ghani et al. "Compositional Game Theory". In: *Proceedings of the 33rd Annual ACM/IEEE Symposium on Logic in Computer Science (LICS)*. Association for Computing Machinery, 2018, pp. 472–481. DOI: 10.1145/3209108.3209165.

[24] Michèle Giry. "A Categorical Approach to Probability Theory". In: *Categorical Aspects of Topology and Analysis*. Vol. 915. Lecture Notes in Mathematics. Springer, 1982, pp. 68–85. DOI: 10.1007/BFb0092872.

[25] David E. Goldberg. *Genetic Algorithms in Search, Optimization, and Machine Learning*. Addison–Wesley, 1989. ISBN: 9780201157673.

[26] Jules Hedges. "Morphisms of Open Games". In: *Electronic Notes in Theoretical Computer Science* 341 (2018), pp. 151–177. DOI: 10.1016/j.entcs.2018.11.008.

[27] Jennifer A. Hoeting et al. "Bayesian Model Averaging: A Tutorial". In: *Statistical Science* 14.4 (1999), pp. 382–417. DOI: 10.1214/ss/1009212519.

[28] John H. Holland. *Adaptation in Natural and Artificial Systems*. University of Michigan Press, 1975.

[29] Pavel Izmailov et al. "Averaging Weights Leads to Wider Optima and Better Generalization". In: *Proceedings of the 34th Conference on Uncertainty in Artificial Intelligence (UAI)*. 2018.

[30] Bart Jacobs. "Probabilities, Distribution Monads, and Convex Categories". In: *Theoretical Computer Science* 412.28 (2011), pp. 3323–3336. DOI: 10.1016/j.tcs.2011.04.005.

[31] Olav Kallenberg. *Foundations of Modern Probability*. 2nd ed. Probability and Its Applications. Springer, 2002. DOI: 10.1007/978-1-4757-4015-8.

[32] Shin-ya Katsumata. "Parametric Effect Monads and Semantics of Effect Systems". In: *Proceedings of the 41st ACM SIGPLAN-SIGACT Symposium on Principles of Programming Languages (POPL '14)*. ACM, 2014, pp. 633–645. DOI: 10.1145/2535838.2535846.

[33] G. M. Kelly. *Basic Concepts of Enriched Category Theory*. Vol. 64. London Mathematical Society Lecture Note Series. Reprinted as *Reprints in Theory and Applications of Categories* No. 10 (2005). Cambridge University Press, 1982.

[34] Victor Klee and George J. Minty. "How Good is the Simplex Algorithm?" In: *Inequalities III*. Ed. by Oved Shisha. Academic Press, 1972, pp. 159–175.

[35] Harold J. Kushner and G. George Yin. *Stochastic Approximation and Recursive Algorithms and Applications*. 2nd ed. Vol. 35. Applications of Mathematics. Springer, 2003. DOI: 10.1007/b97441.

[36] Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. "Simple and Scalable Predictive Uncertainty Estimation using Deep Ensembles". In: *Advances in Neural Information Processing Systems 30*. 2017.

[37] F. William Lawvere. "Metric Spaces, Generalized Logic, and Closed Categories". In: *Reprints in Theory and Applications of Categories* 1 (2002). Reprint of the 1973 article in *Rendiconti del Seminario Matematico e Fisico di Milano* 43 (1973), 135–166, pp. 1–37.

[38] Tom Leinster. *Higher Operads, Higher Categories*. Vol. 298. London Mathematical Society Lecture Note Series. Cambridge University Press, 2004.

[39] Xavier Leroy. "Formal Verification of a Realistic Compiler". In: *Communications of the ACM* 52.7 (2009), pp. 107–115. DOI: 10.1145/1538788.1538814.

[40] Sheng Liang, Paul Hudak, and Mark P. Jones. "Monad Transformers and Modular Interpreters". In: *Conference Record of POPL '95: 22nd ACM SIGPLAN-SIGACT Symposium on Principles of Programming Languages*. ACM, 1995, pp. 333–343. DOI: 10.1145/199448.199528.

[41] Torgny Lindvall. *Lectures on the Coupling Method*. Wiley-Interscience, 1992. ISBN: 0471540250.

[42] Jacob Lurie. *Higher Topos Theory*. Vol. 170. Annals of Mathematics Studies. Princeton University Press, 2009.

[43] Jacob Lurie. *Higher Algebra*. Available online as a preprint. 2017.

[44] Saunders Mac Lane. *Categories for the Working Mathematician*. 2nd ed. Vol. 5. Graduate Texts in Mathematics. Springer, 1998. DOI: 10.1007/978-1-4757-4721-8.

[45] Eugenio Moggi. "Notions of Computation and Monads". In: *Information and Computation* 93.1 (1991), pp. 55–92. DOI: 10.1016/0890-5401(91)90052-4.

[46] Dov Monderer and Lloyd S. Shapley. "Potential Games". In: *Games and Economic Behavior* 14.1 (1996), pp. 124–143. DOI: 10.1006/game.1996.0044.

[47] John Nash. "Non-Cooperative Games". In: *Annals of Mathematics* 54.2 (1951), pp. 286–295. URL: https://www.jstor.org/stable/1969529.

[48] John F. Nash. "Equilibrium Points in n-Person Games". In: *Proceedings of the National Academy of Sciences of the United States of America* 36.1 (1950), pp. 48–49. DOI: 10.1073/pnas.36.1.48.

[49] Yurii Nesterov. *Introductory Lectures on Convex Optimization: A Basic Course*. Kluwer Academic Publishers, 2004. ISBN: 9781402075537. DOI: 10.1007/978-1-4419-8853-9.

[50] Yurii Nesterov and Arkadii Nemirovskii. *Interior-Point Polynomial Algorithms in Convex Programming*. SIAM, 1994. DOI: 10.1137/1.9781611970791.

[51] John von Neumann and Oskar Morgenstern. *Theory of Games and Economic Behavior*. 60th Anniversary Edition. Princeton University Press, 2007. ISBN: 9780691130613.

[52] Hukukane Nikaidō and Kyosi Isoda. "Note on Noncooperative Convex Games". In: *Pacific Journal of Mathematics* 5.5 (1955), pp. 807–815.

[53] Jorge Nocedal and Stephen J. Wright. *Numerical Optimization*. 2nd ed. Springer, 2006. DOI: 10.1007/978-0-387-40065-5.

[54] Bernt Øksendal. *Stochastic Differential Equations: An Introduction with Applications*. 6th ed. Universitext. Springer, 2003. DOI: 10.1007/978-3-642-14394-6.

[55] Dominic Orchard, Philip Wadler, and Harley Eades III. "Unifying Graded and Parameterised Monads". In: *Proceedings of the Eighth Workshop on Mathematically Structured Functional Programming (MSFP 2020)*. Vol. 317. Electronic Proceedings in Theoretical Computer Science. 2020, pp. 18–38. DOI: 10.4204/EPTCS.317.2.

[56] Martin J. Osborne and Ariel Rubinstein. *A Course in Game Theory*. MIT Press, 1994. ISBN: 9780262150415.

[57] Christos H. Papadimitriou. *Computational Complexity*. Addison–Wesley, 1994. ISBN: 9780201530827.

[58] Christos H. Papadimitriou and Mihalis Yannakakis. "Optimization, Approximation, and Complexity Classes". In: *Journal of Computer and System Sciences* 43.3 (1991), pp. 425–440. DOI: 10.1016/0022-0000(91)90023-X.

[59] Charles Rezk. "A Model for the Homotopy Theory of Homotopy Theory". In: *Transactions of the American Mathematical Society* 353.3 (2001), pp. 973–1007. DOI: 10.1090/S0002-9947-00-02653-2.

[60] Charles Rezk. "A Cartesian Presentation of Weak n-Categories". In: *Geometry & Topology* 14.1 (2010), pp. 521–571. DOI: 10.2140/gt.2010.14.521.

[61] Emily Riehl and Dominic Verity. *Elements of ∞-Category Theory*. Available as a preprint in earlier versions. Cambridge University Press, 2022.

[62] Herbert Robbins and Sutton Monro. "A Stochastic Approximation Method". In: *The Annals of Mathematical Statistics* 22.3 (1951), pp. 400–407. DOI: 10.1214/aoms/1177729586.

[63] Christian P. Robert and George Casella. *Monte Carlo Statistical Methods*. 2nd ed. Springer Texts in Statistics. Springer, 2004. DOI: 10.1007/978-1-4757-4145-2.

[64] R. Tyrrell Rockafellar and Roger J.-B. Wets. *Variational Analysis*. Vol. 317. Grundlehren der mathematischen Wissenschaften. Springer, 1998. DOI: 10.1007/978-3-642-02431-3.

[65] J. B. Rosen. "Existence and Uniqueness of Equilibrium Points for Concave N-Person Games". In: *Econometrica* 33.3 (1965), pp. 520–534. URL: https://www.jstor.org/stable/1911749.

[66] Håvard Rue and Leonhard Held. *Gaussian Markov Random Fields: Theory and Applications*. Monographs on Statistics and Applied Probability. Chapman & Hall/CRC, 2005. ISBN: 9781584884323.

[67] Alexander Shapiro, Darinka Dentcheva, and Andrzej Ruszczyński. *Lectures on Stochastic Programming: Modeling and Theory*. 2nd ed. SIAM, 2014. DOI: 10.1137/1.9781611973433.

[68] Lloyd S. Shapley. "Stochastic Games". In: *Proceedings of the National Academy of Sciences of the United States of America* 39.10 (1953), pp. 1095–1100. DOI: 10.1073/pnas.39.10.1095.

[69] Maurice Sion. "On General Minimax Theorems". In: *Pacific Journal of Mathematics* 8.1 (1958), pp. 171–176. DOI: 10.2140/pjm.1958.8.171.

[70] Ross Street. "The Formal Theory of Monads". In: *Journal of Pure and Applied Algebra* 2.2 (1972), pp. 149–168. DOI: 10.1016/0022-4049(72)90019-9.

[71] Vijay V. Vazirani. *Approximation Algorithms*. Springer, 2001. DOI: 10.1007/978-3-662-04565-7.

[72] Philip Wadler. "Monads for Functional Programming". In: *Advanced Functional Programming*. Ed. by Johan Jeuring and Erik Meijer. Vol. 925. Lecture Notes in Computer Science. Springer, 1995, pp. 24–52. DOI: 10.1007/3-540-59451-5_2.

[73] David H. Wolpert. "Stacked Generalization". In: *Neural Networks* 5.2 (1992), pp. 241–259. DOI: 10.1016/S0893-6080(05)80023-1.